Burning and Suppression of Solids (BASS): An Overview of Recent ISS Results





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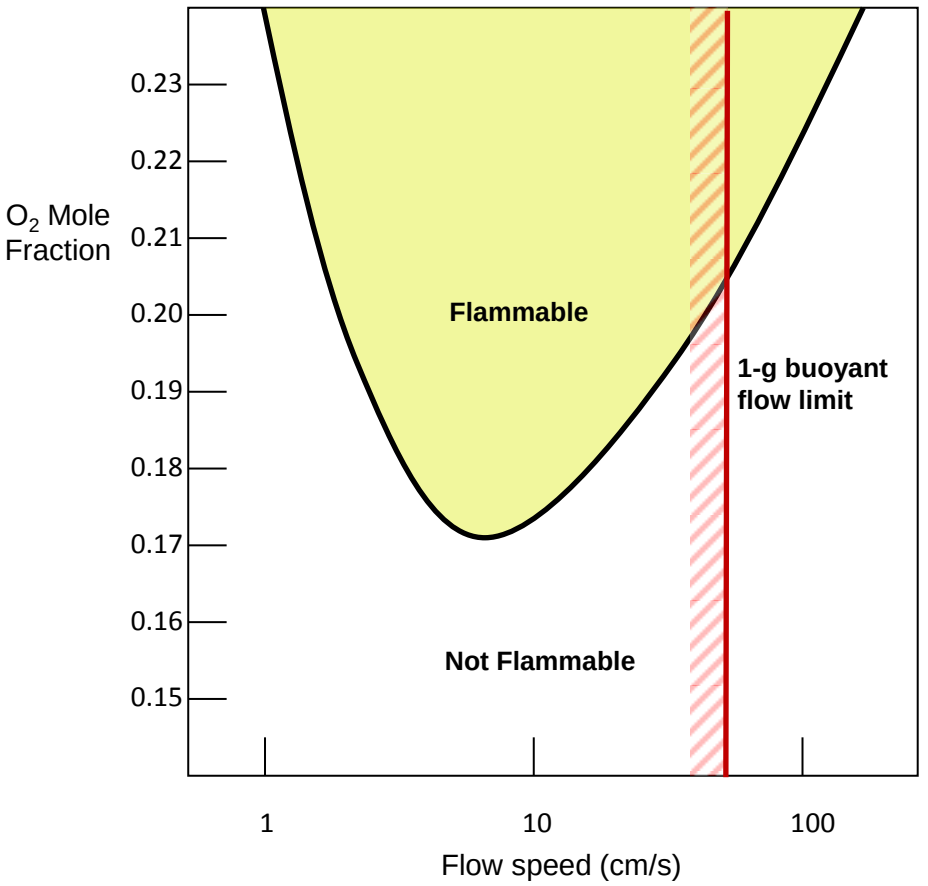
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Science Background:

- Combustion of solid fuels in microgravity has been an active area of research
- Numerical models predict that 1-g buoyant flows profoundly affect fuel flammability
- Ground-based facilities have verified this behavior but only for thin fuels in opposed flow
- Flow speeds below the buoyant limit are of interest

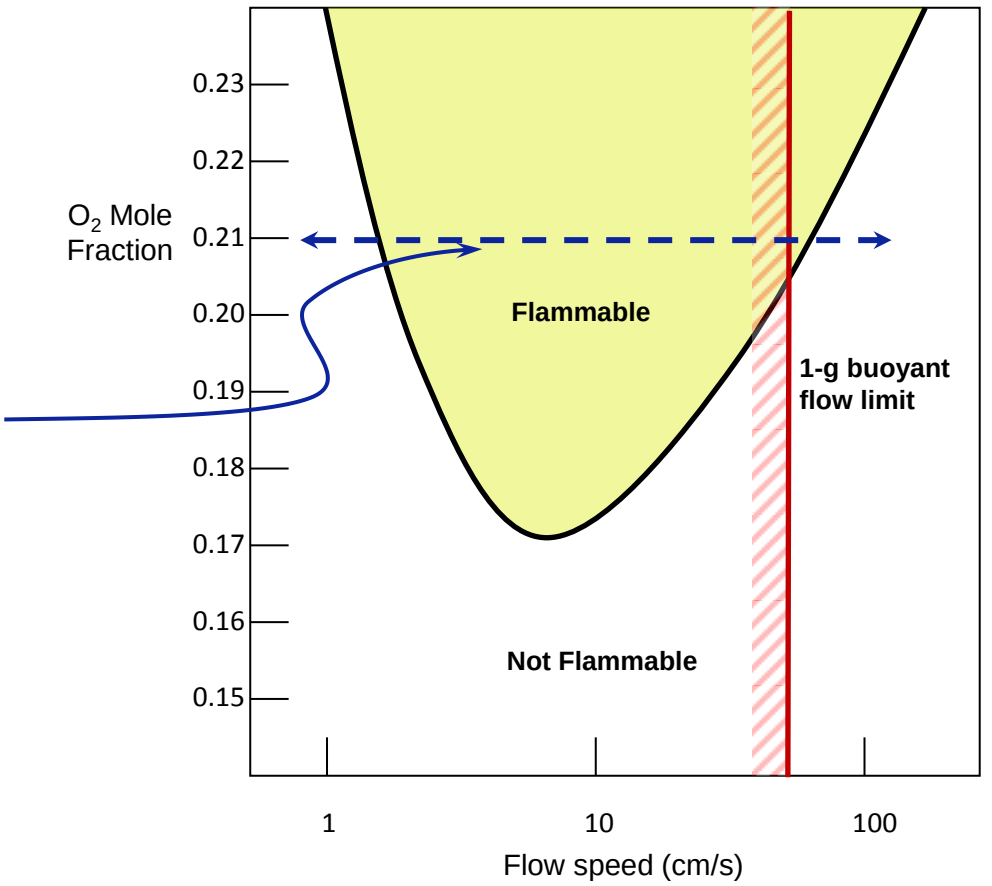
Typical Flammability Boundary for a Solid Fuel



Hypotheses:

- Materials in microgravity, with adequate ventilation, may burn more readily compared to normal gravity
- Testing at constant O_2 mole fraction permits studying flame spread and extinction at low and high flow speed.

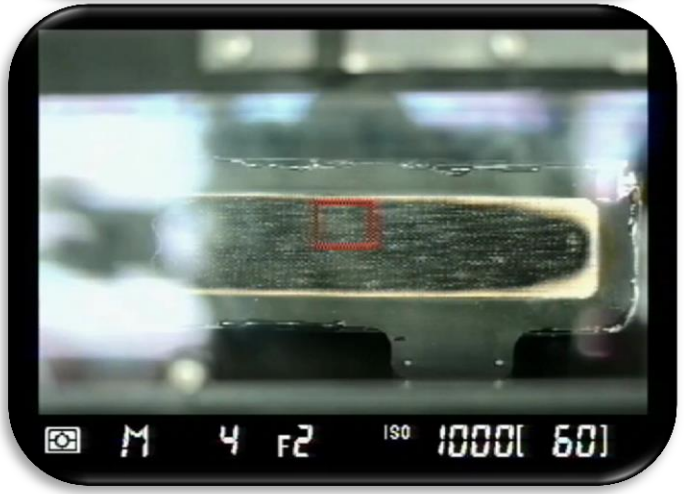
Typical Flammability Boundary for a Solid Fuel



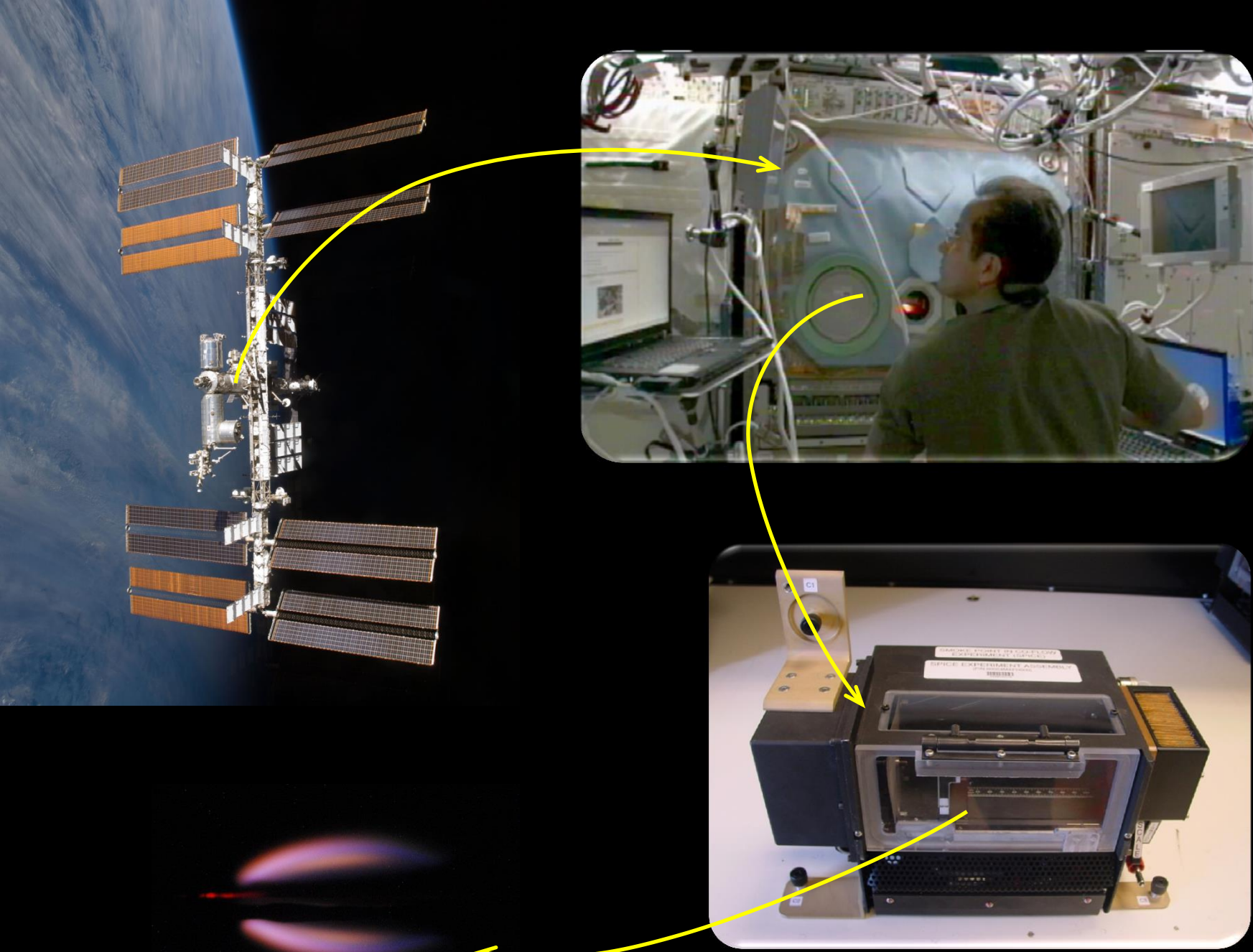
Approach:

- Microgravity combustion tests were performed aboard the International Space Station using the BASS (Burning and Suppression of Solids) hardware.
- The wind tunnel was installed in the Microgravity Science Glovebox which supplied power, imaging, and a level of containment.
- Fuel samples were mounted inside a small wind tunnel which could impose airflow speeds up to 40 cm/s.
- The effects of airflow speed on flame appearance, flame growth, and spread rates were determined in both the opposed and concurrent flow.
- More recent tests varying oxygen (below 21%) have been completed.



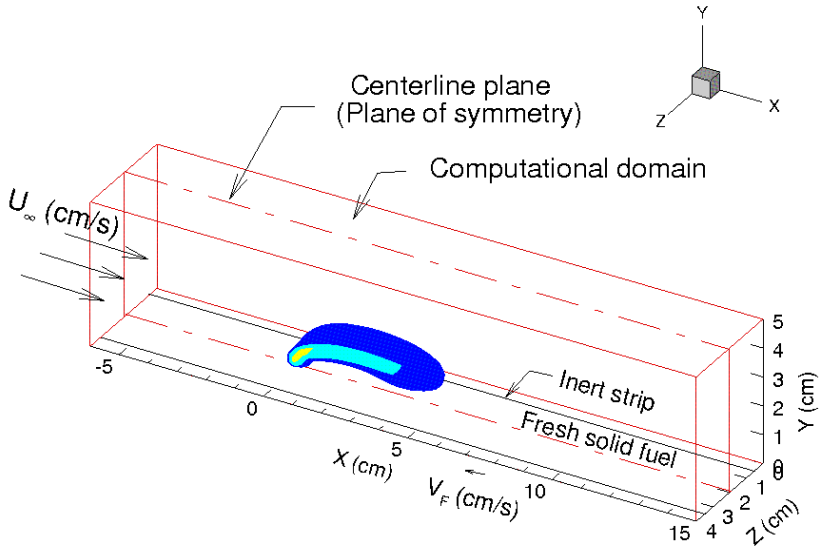


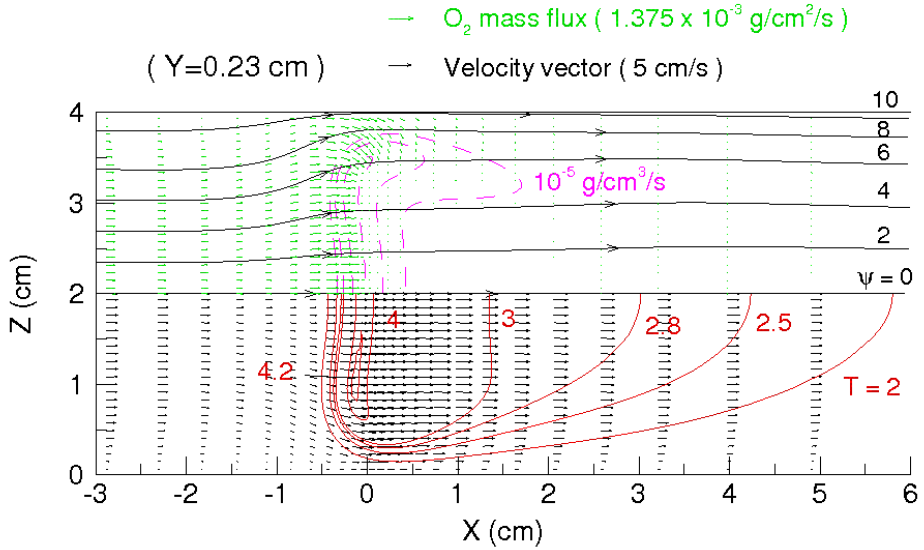




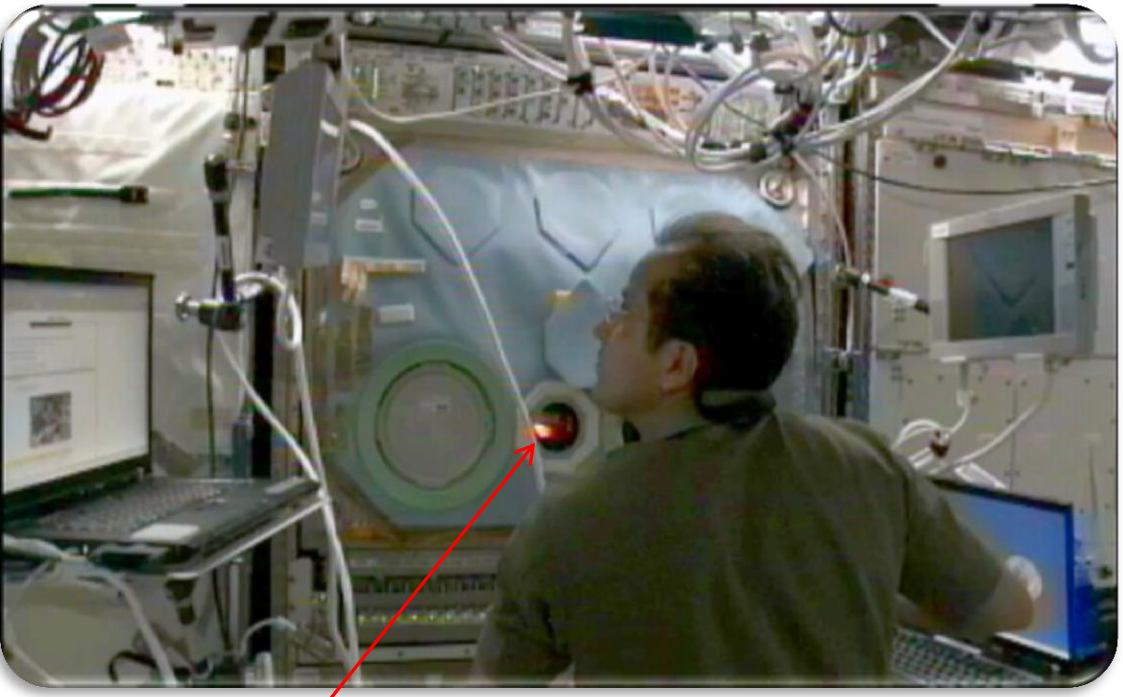
Science Applications:

- Understanding of long-duration microgravity solid material burning and extinction
- Improved strategies for NASA spacecraft materials selection; link actual burn behavior in microgravity to Earth-based selection methods
- Improved combustion computational models used in the design of fire detection and suppression systems in microgravity and on Earth
- Validated detailed combustion models in the simpler flow environment of microgravity build more complex combustion models needed to capture the important details of flames burning in normal gravity; models have wide applicability to the general understanding of many terrestrial combustion problems.

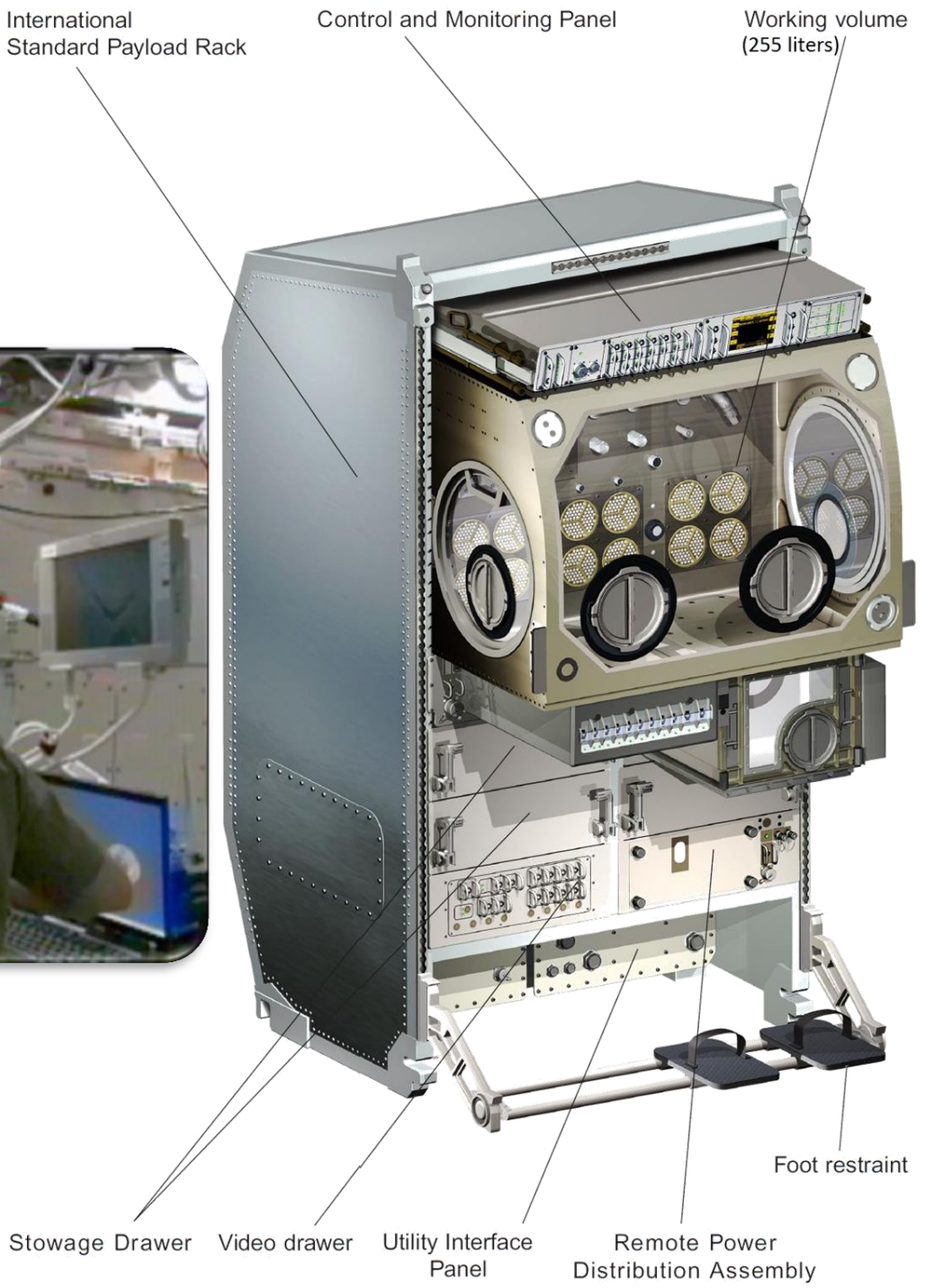


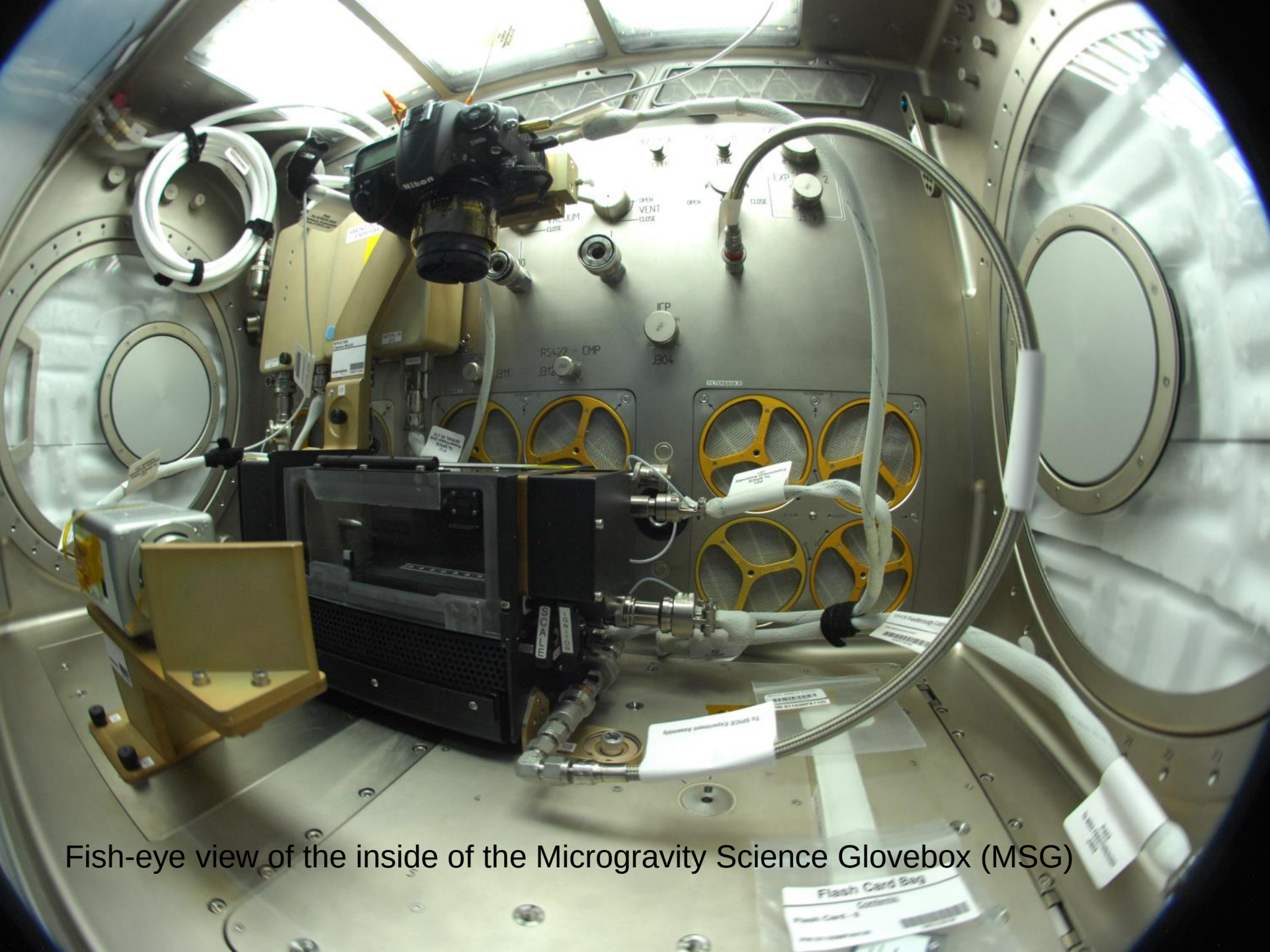


Microgravity Science Glovebox (MSG)



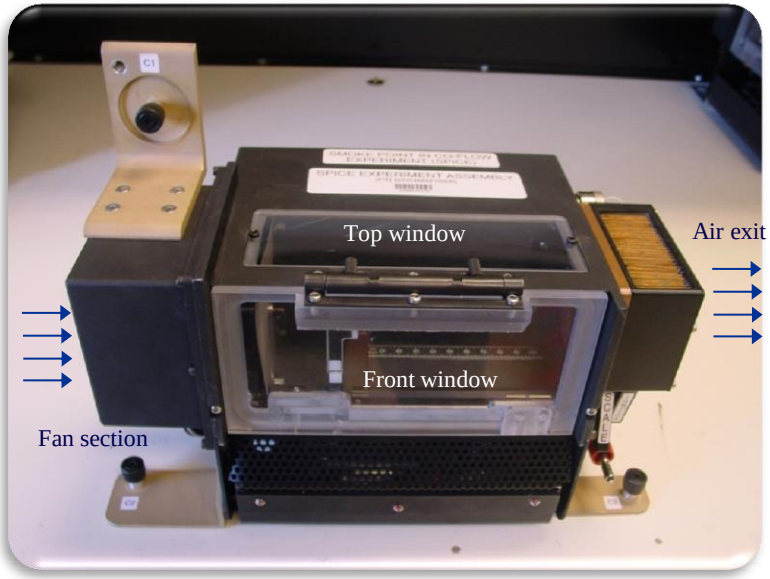
BASS flame visible through viewport



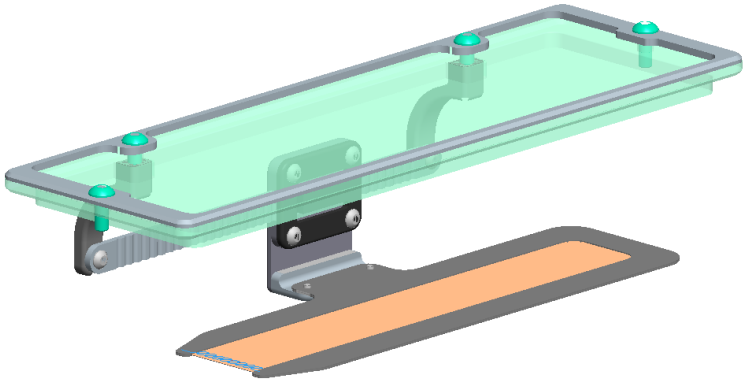


Fish-eye view of the inside of the Microgravity Science Glovebox (MSG)

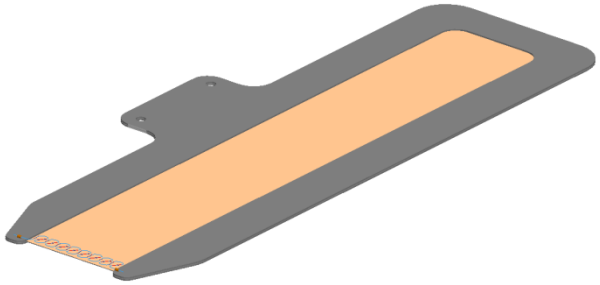
Hardware Details



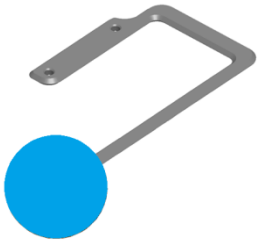
Top window



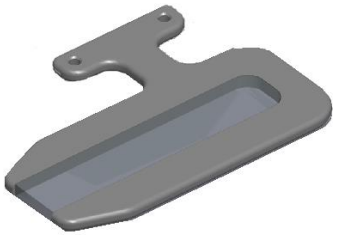
Permits variety of solid samples to be mounted, ignited, and burned:



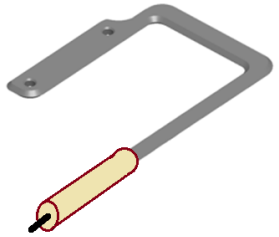
Thin, flat (17)



Spheres (12)



Thick, flat (4)

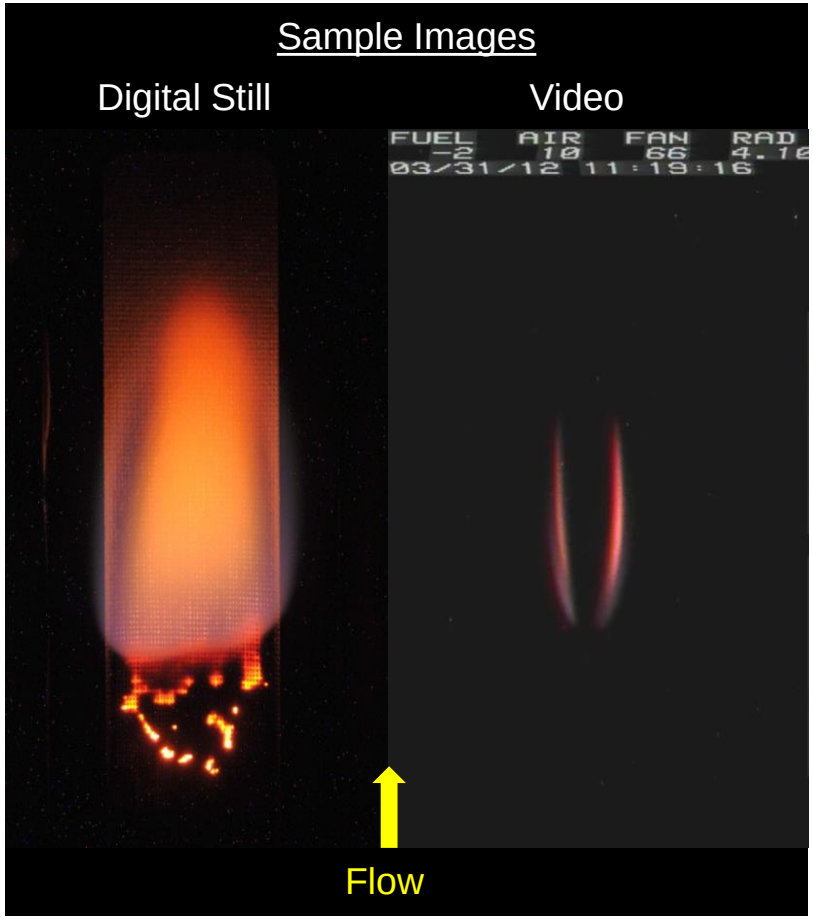


Candles (8)

(Note: samples are not to scale; samples can be flipped 180 degrees if desired)

Hardware Summary:

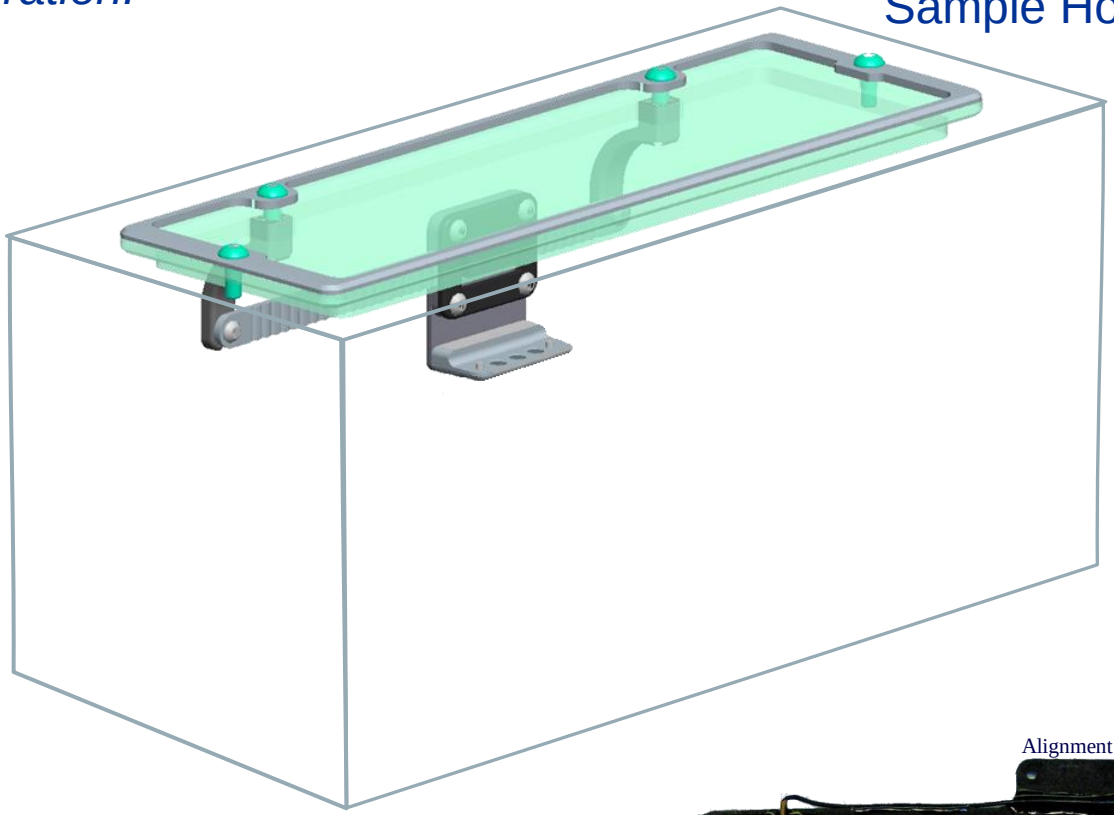
- BASS utilized the on-orbit SPICE hardware; minor modifications were made to burn solid samples
- Small flow tunnel
- Solid samples were installed, ignited, extinguished, and recorded
- Video and digital still camera provided bulk of the data. Flame appearance, behavior, spread rate, and extinction dynamics were measured
- Airflow speed was the main variable
- 41 samples, 115 burns completed



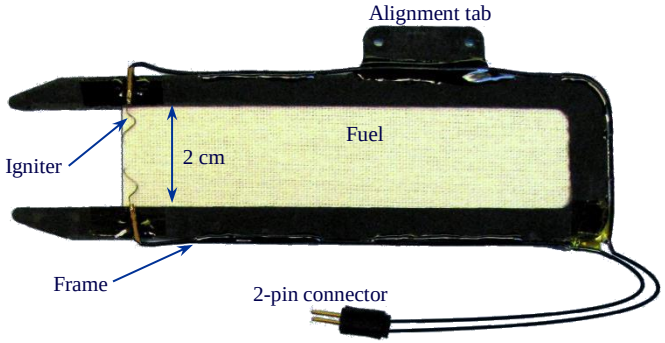
Experiment Operation:

Top Window and Sample Holder

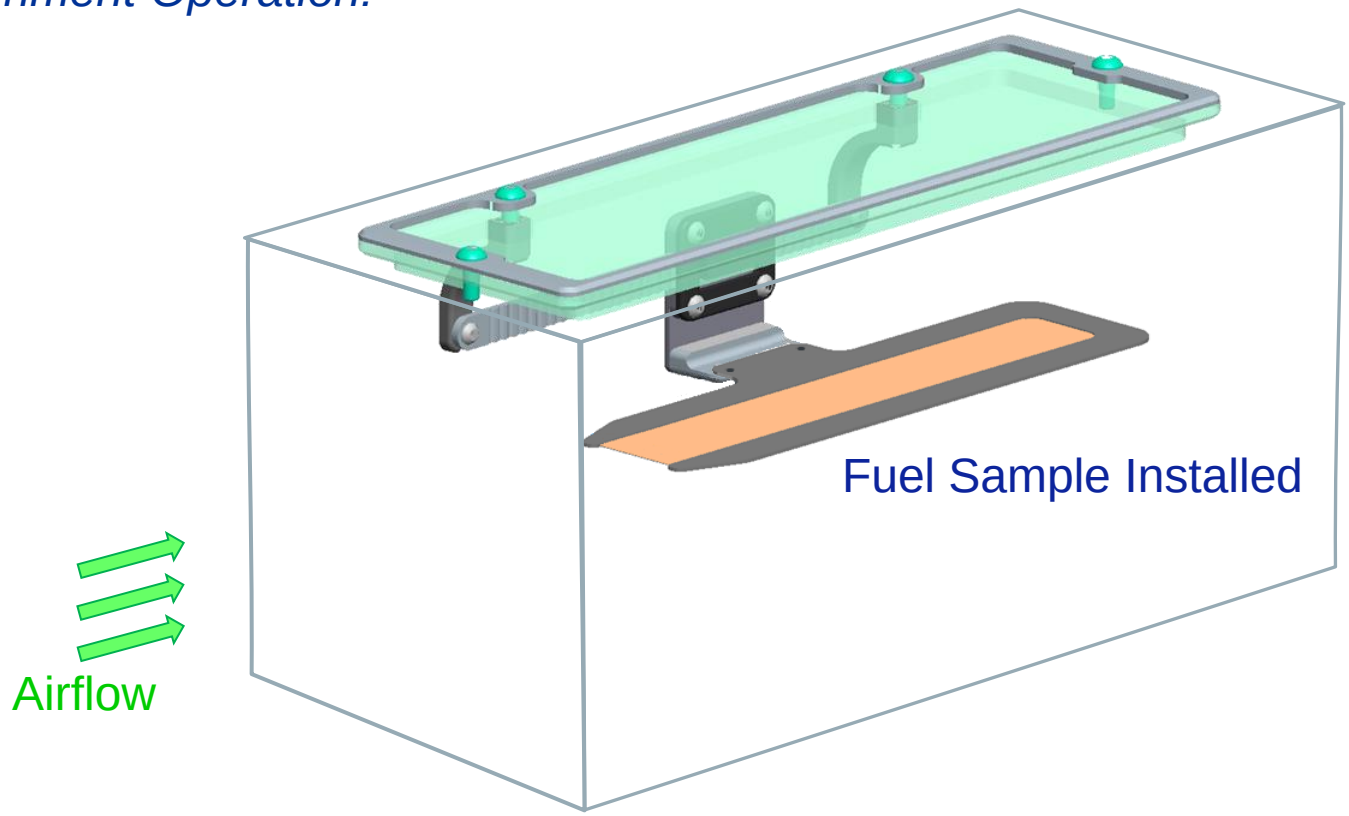
Flow Duct



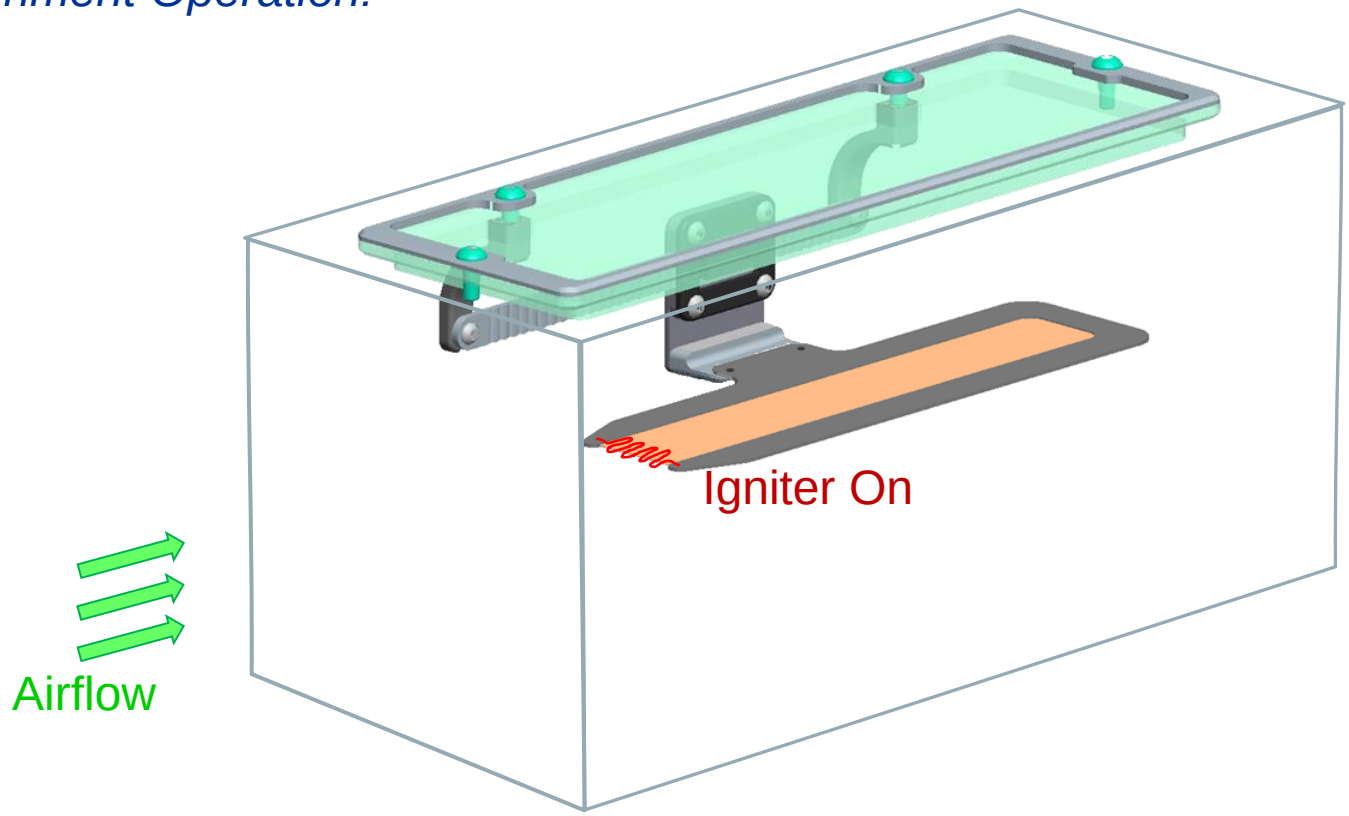
Airflow



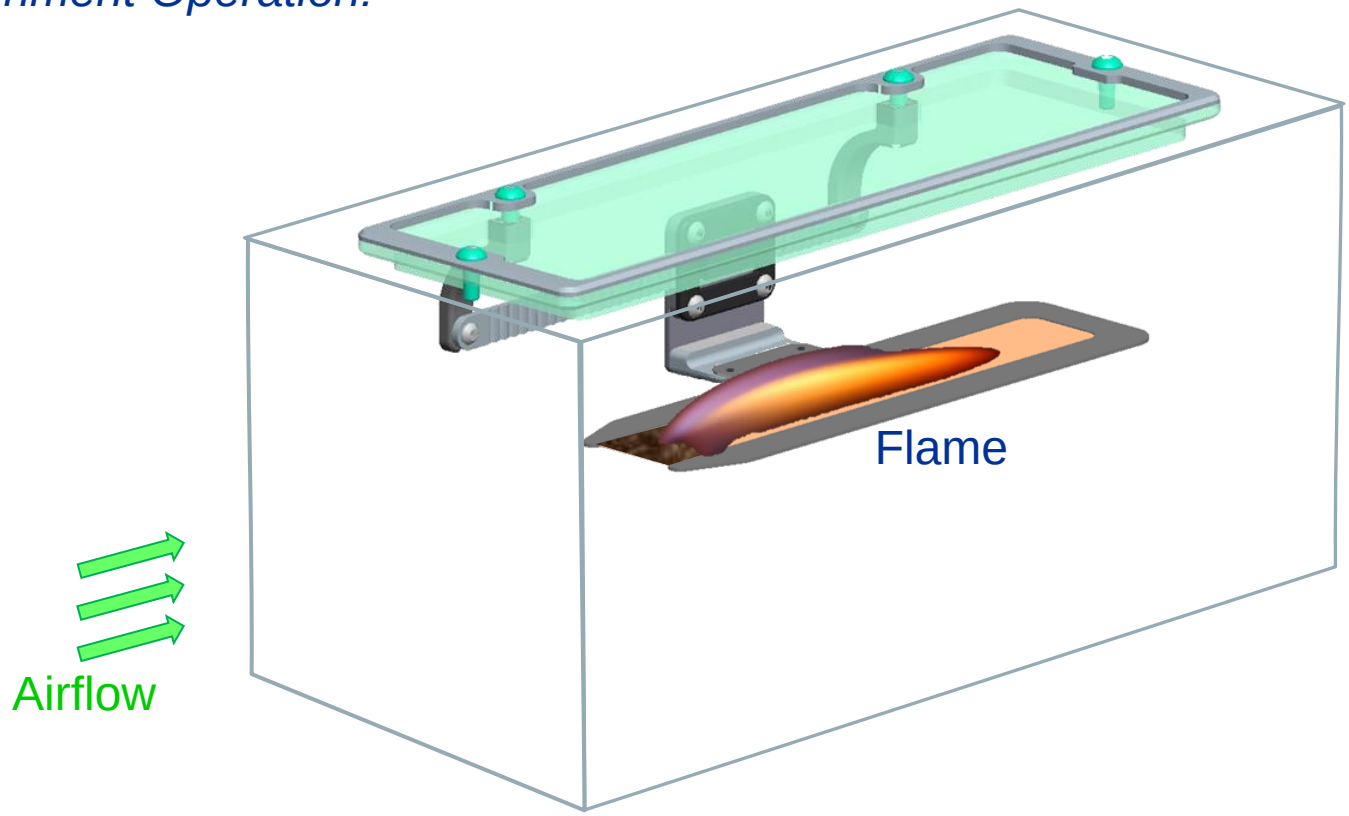
Experiment Operation:



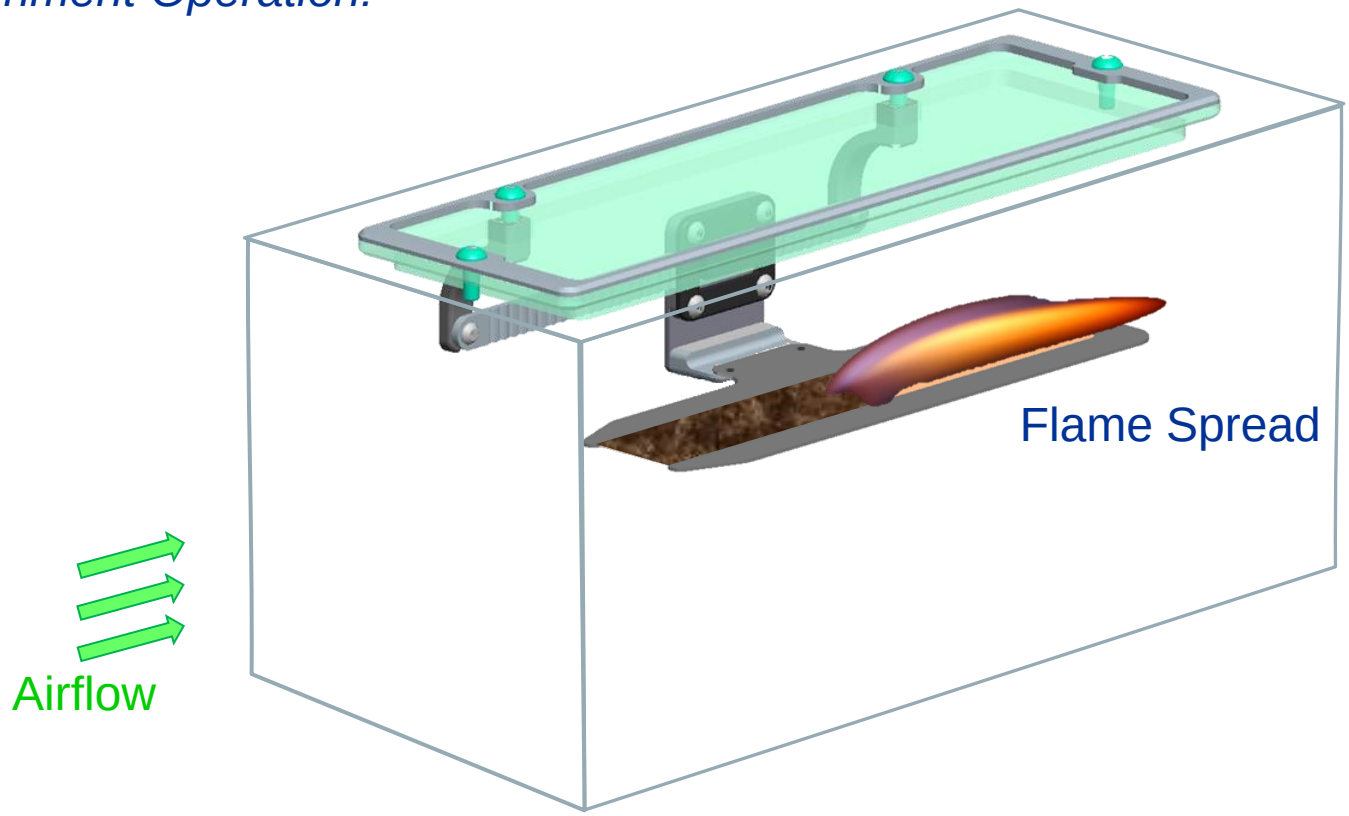
Experiment Operation:



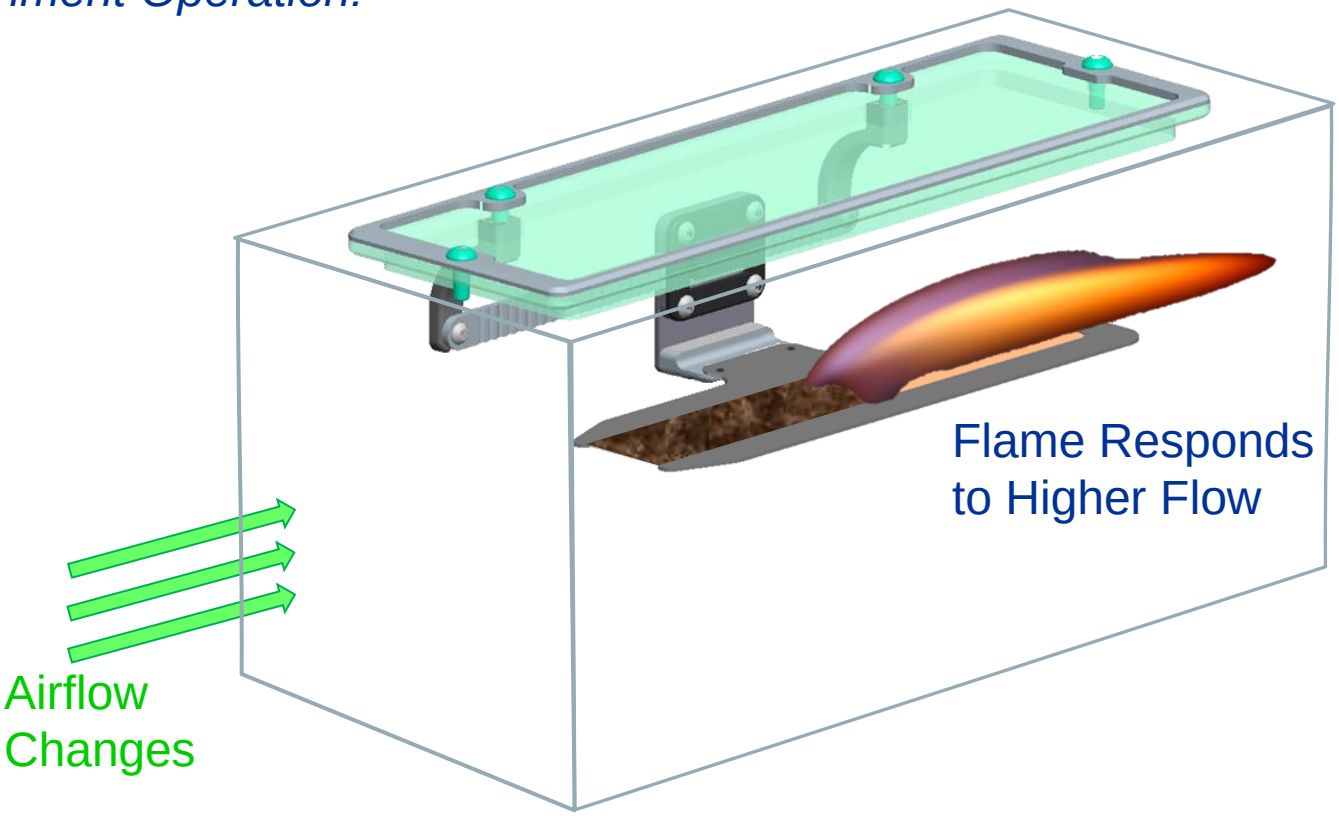
Experiment Operation:



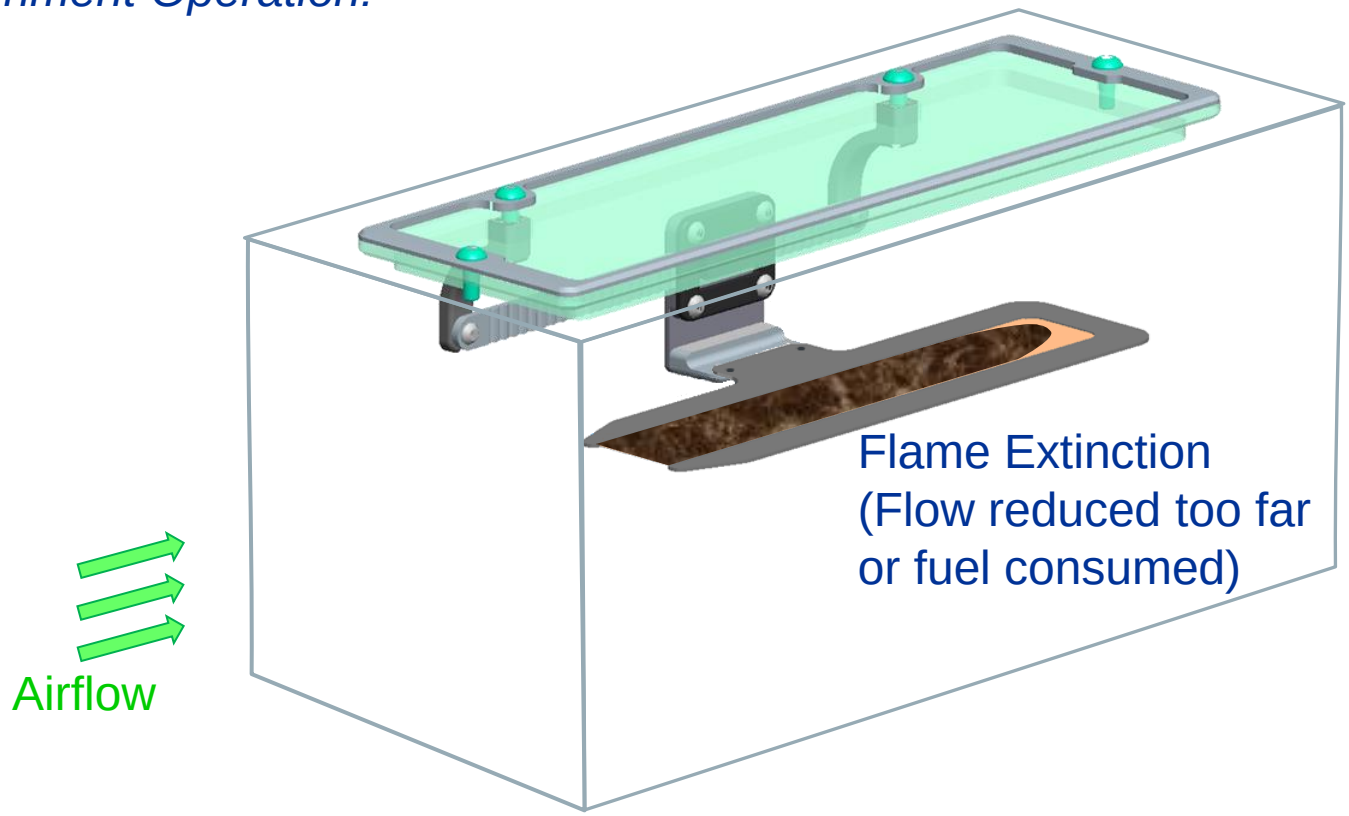
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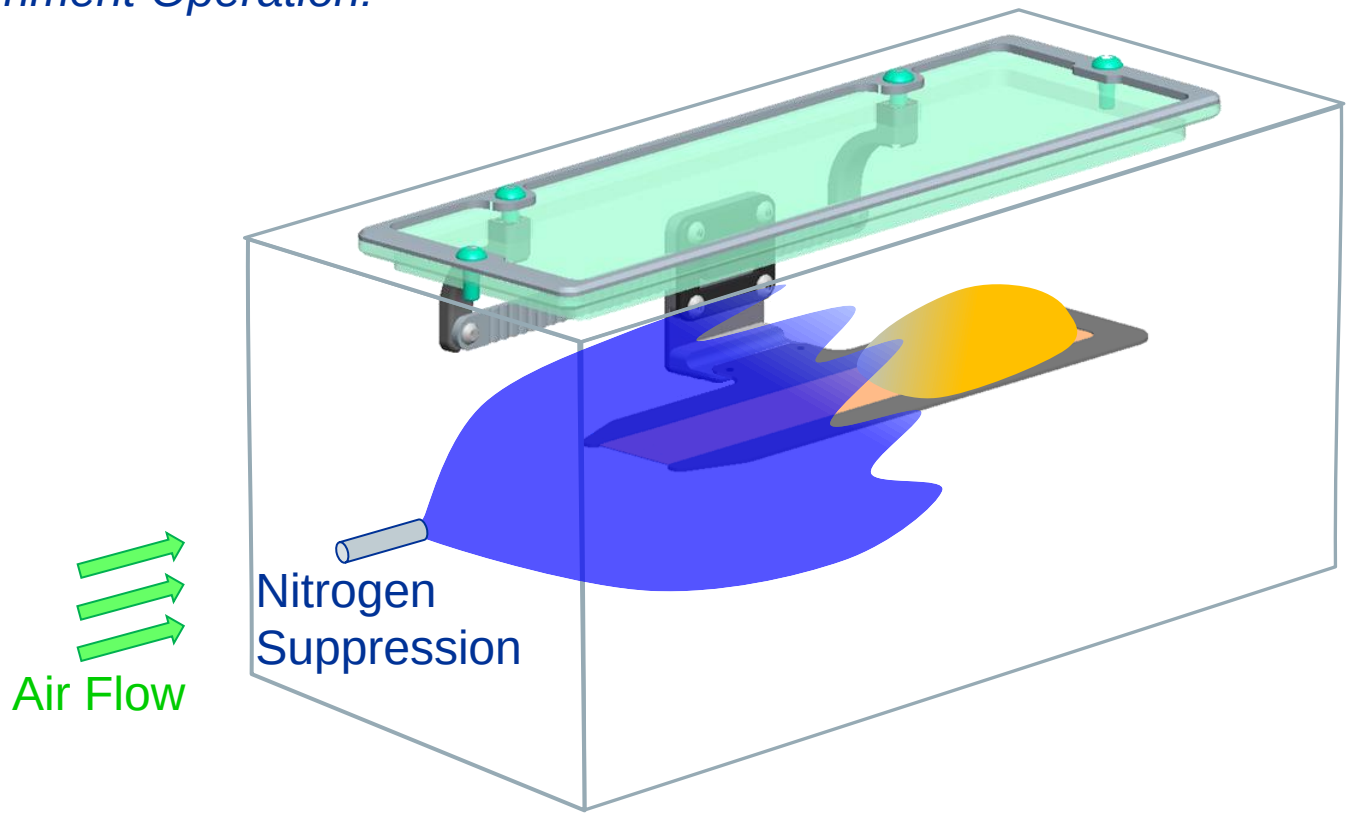
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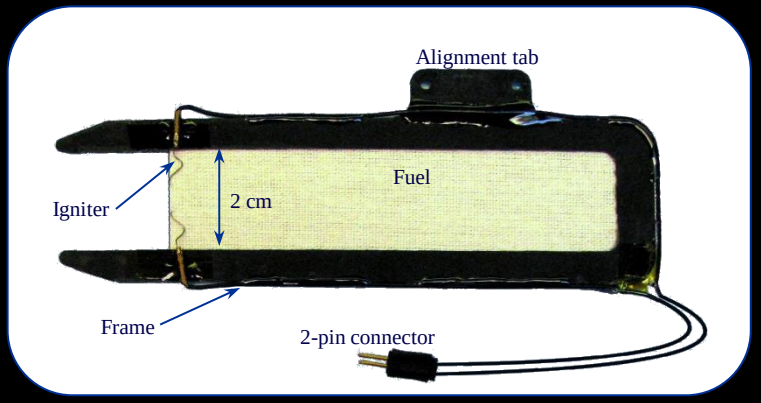
Experiment Operation:

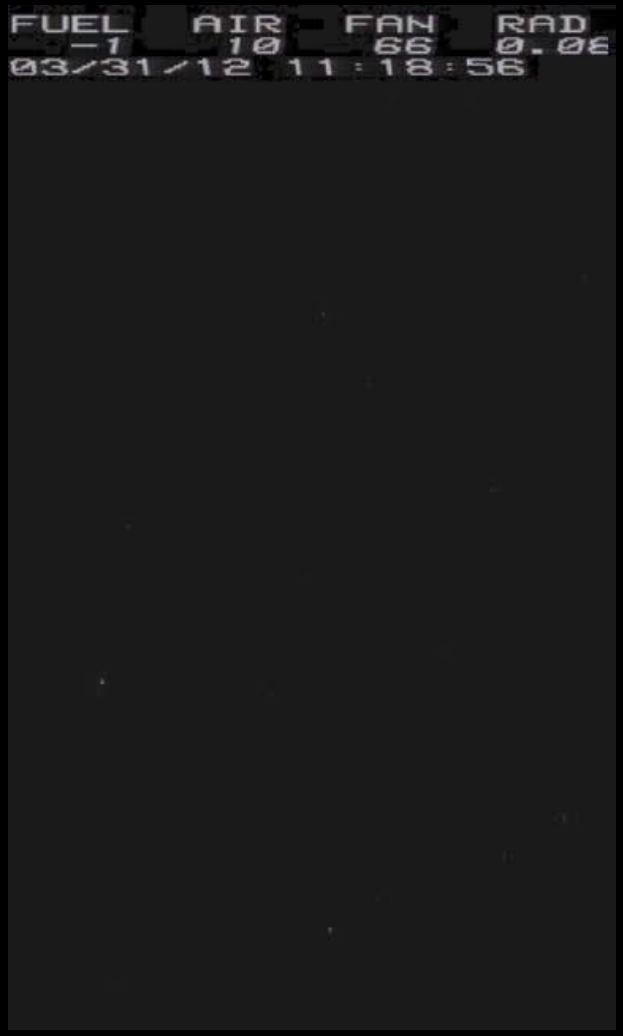


Experiment Operation:



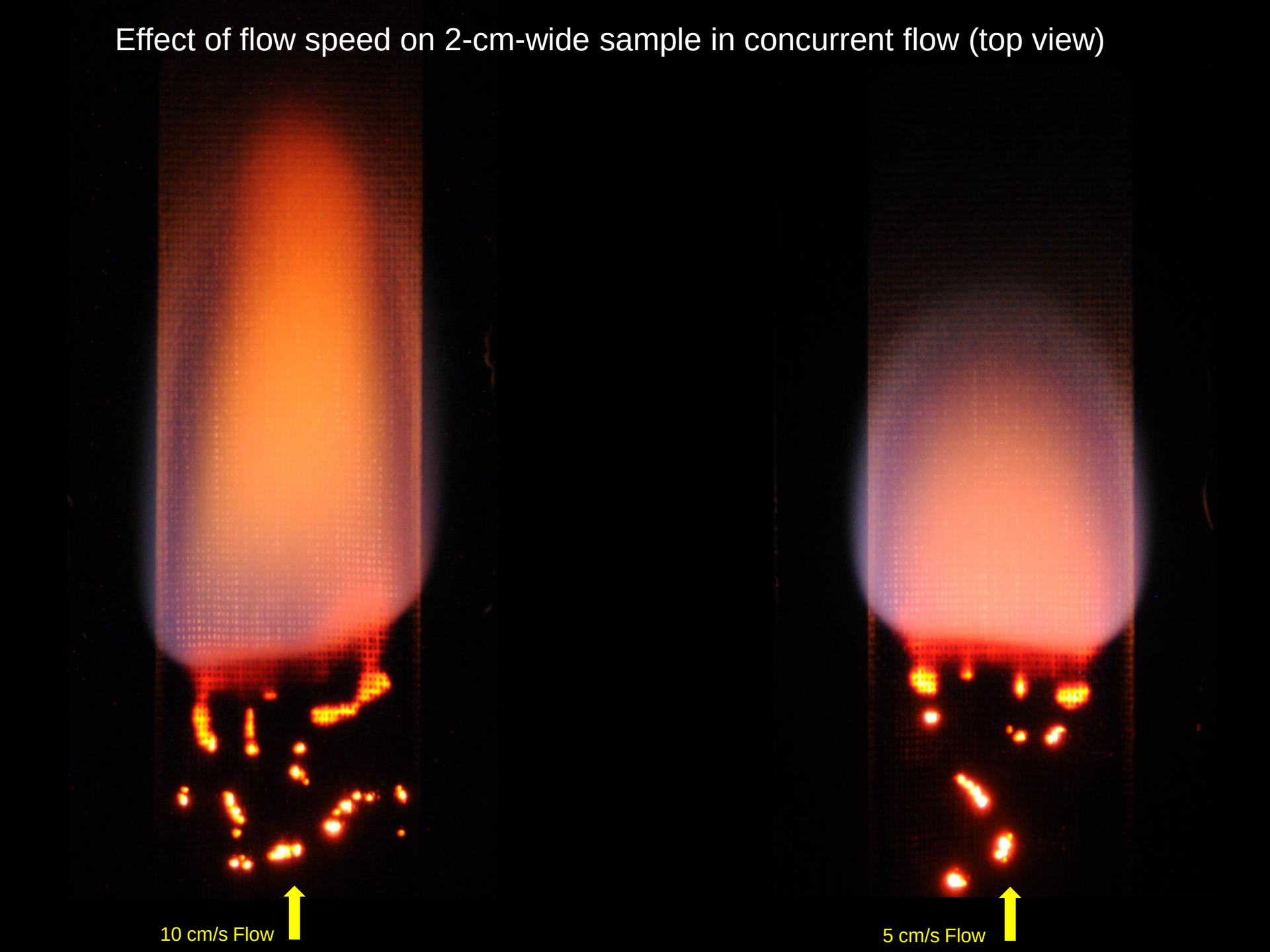
1. Thin Fuel: Flame Spread and Extinction (Cotton / fiberglass fabric)





Video (edge view) of 2-cm-wide sample in concurrent flow.
Ignition at airflow speed of 10 cm/s, then flow is reduced to 5 cm/s.

Effect of flow speed on 2-cm-wide sample in concurrent flow (top view)



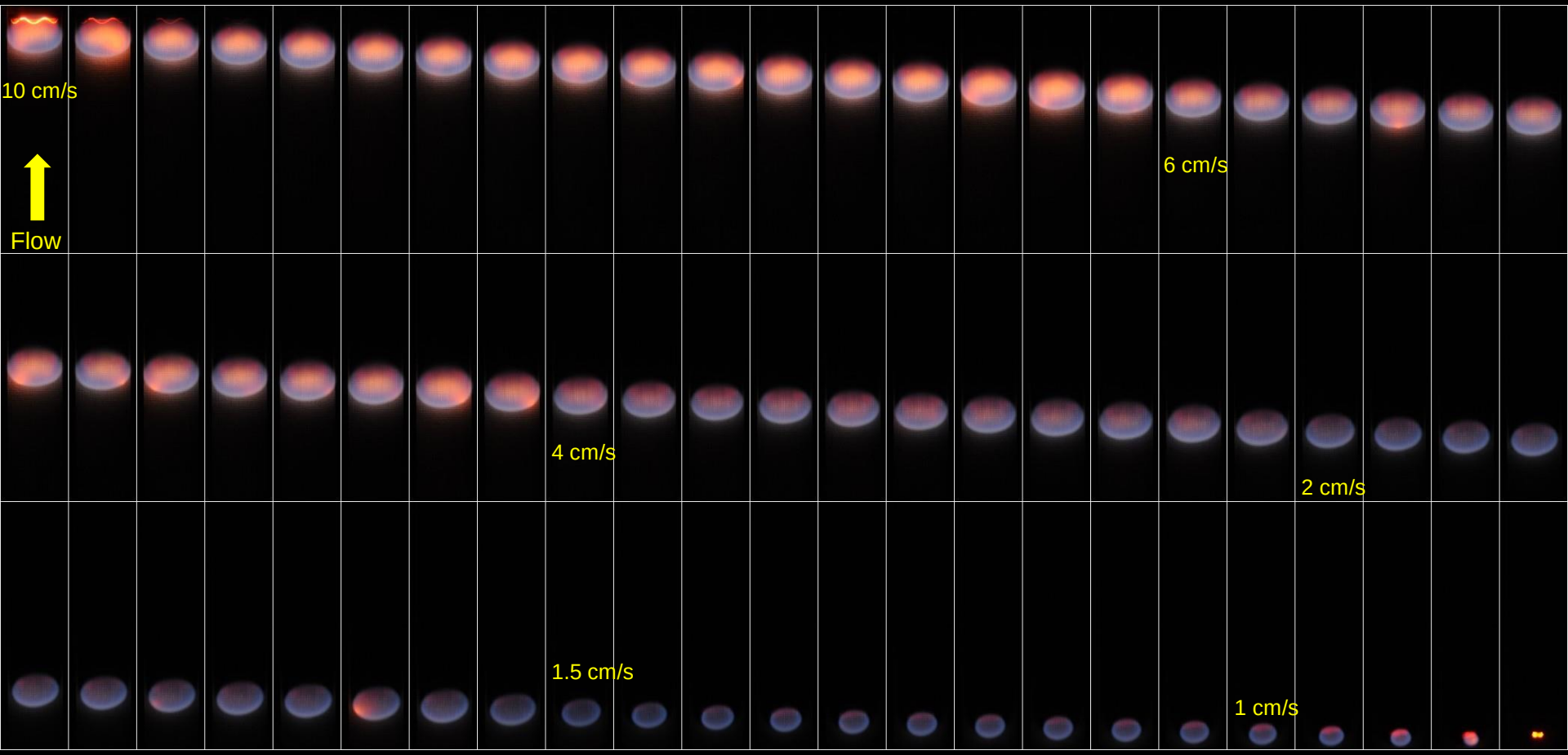
Effect of flow speed on sample in concurrent flow (edge view)



2 cm wide, 20 cm/s flow

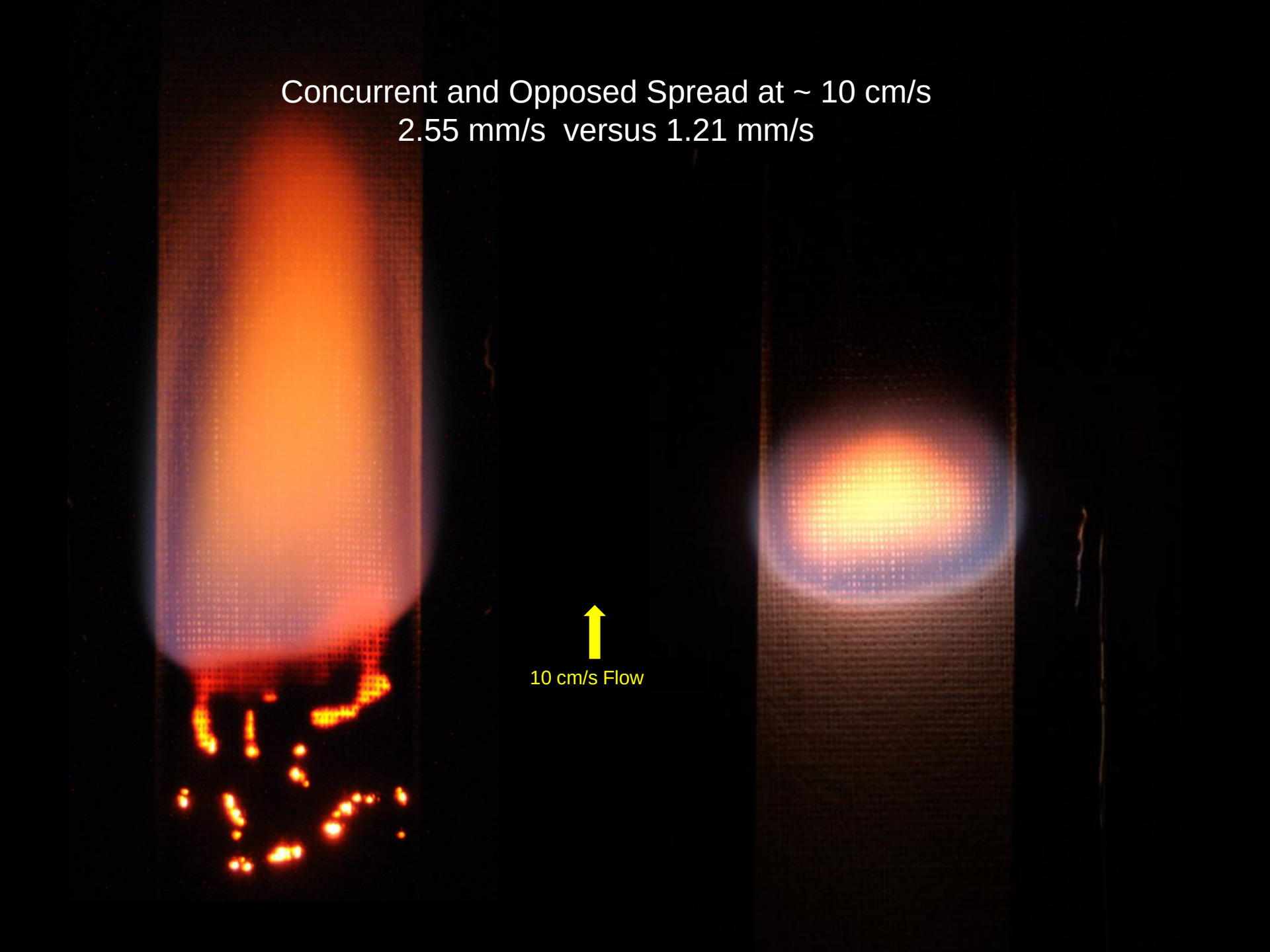


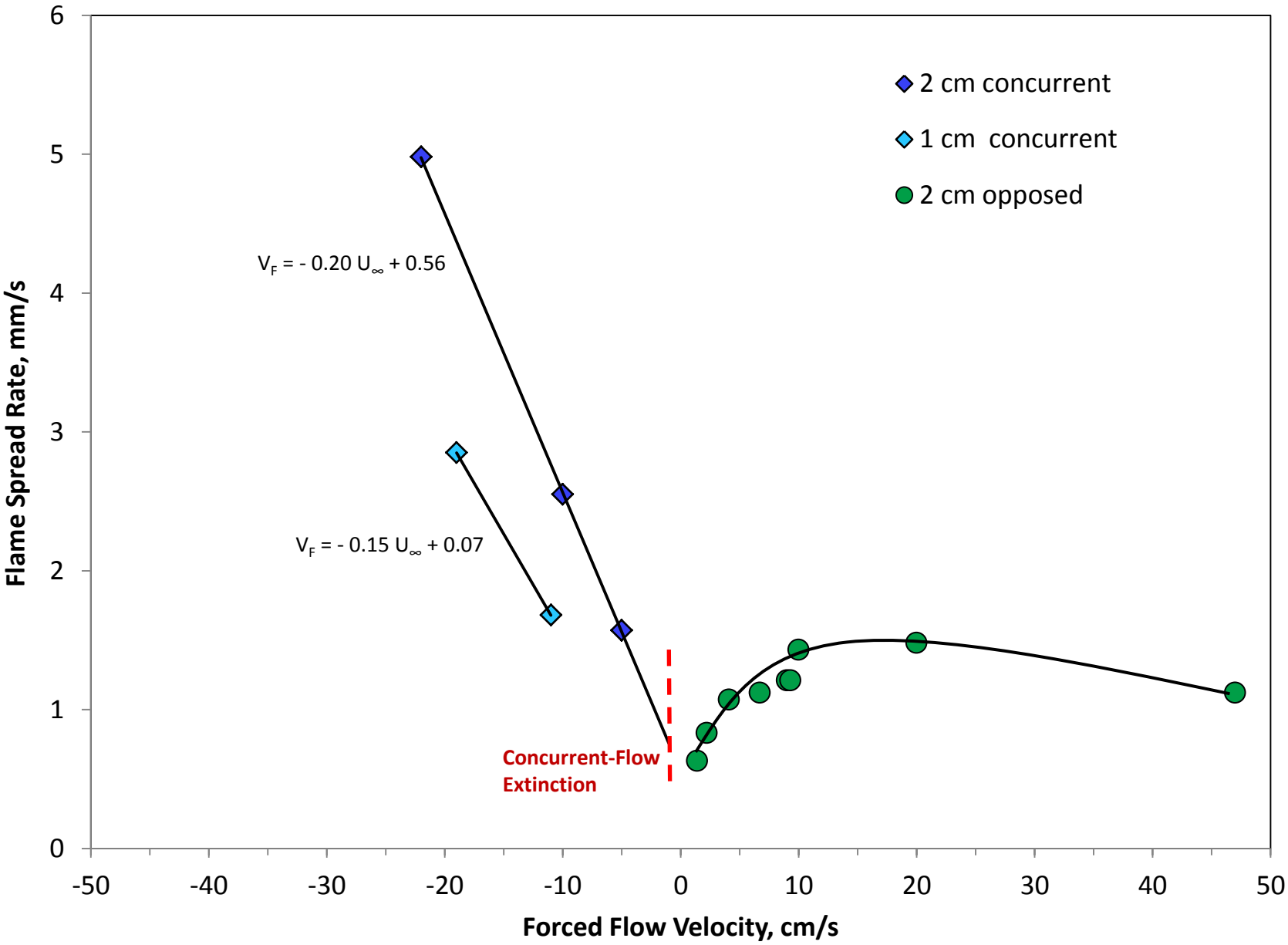
1 cm wide, 10 cm/s flow



Digital still camera images showing a flame burning a 2-cm wide cotton-fiberglass fabric in opposed flow. Images are taken every 1.25 sec (starting at top and moving from left to right). The flow is decreased in discrete steps from 10 cm/s all the way down to about 1 cm/s. The flame response to flow changes is very rapid, and the flow effects on the flame and its spread rate are dramatic. Total burn time is 90 sec. Flow changes are indicated by numbers.

Concurrent and Opposed Spread at ~ 10 cm/s
2.55 mm/s versus 1.21 mm/s




2 cm concurrent
1 cm concurrent
2 cm opposed
$V_F = - 0.20\ U_\infty + 0.56$
$V_F = - 0.15\ U_\infty + 0.07$
Concurrent-Flow Extinction
Flame Spread Rate, mm/s
Forced Flow Velocity, cm/s
0
1
2
3
4
5
6
-50
-40
-30
-20
-10
0
10
20
30
40
50

Summary 1. Thin Fuel: Flame Spread and Extinction

- Opposed flow:
  - flame quickly reached steady spread
  - spread rate was fastest at an intermediate value of flow speed

- Concurrent flow:
  - Flame spread rate increased linearly with increasing flow.
  - Quenching extinction was observed (around 1 cm/s)

- This is the first time that detailed transient flame growth data was obtained in purely forced flows in microgravity.

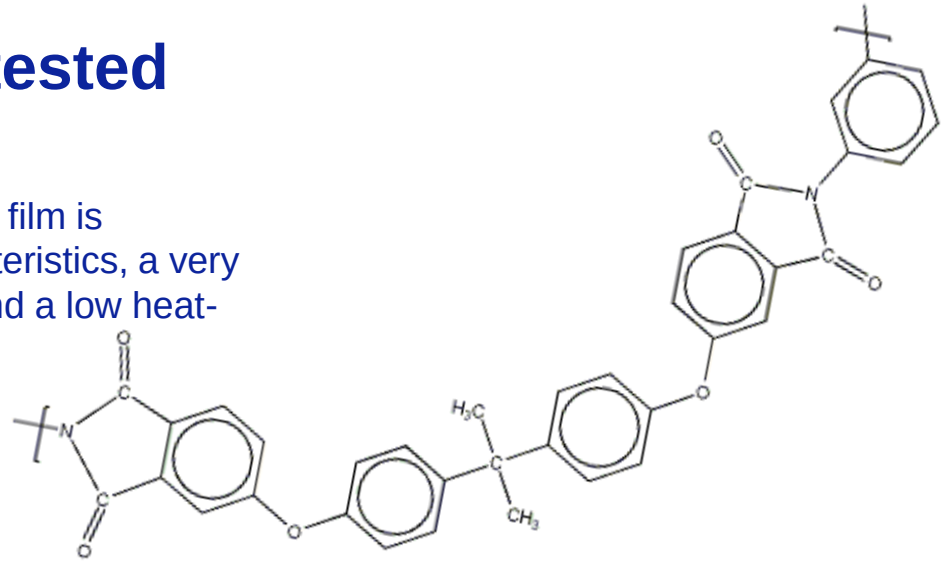
- Long-duration experiments validate a number of theoretical predictions and also provide the data for a transient flame growth model under development.

2. Material Flammability Comparison: 0-g / 1-g

Fuels tested

- **Ultem® 1000** $C_{37}H_{24}O_6N_2$
- Fire retarded polyetherimide (PEI) in 10 mil thick film is inherently flame-retarding, with ***charring*** characteristics, a very low smoke signature, very low smoke toxicity, and a low heat-release rate.

  ULOI: 23.5% O_2



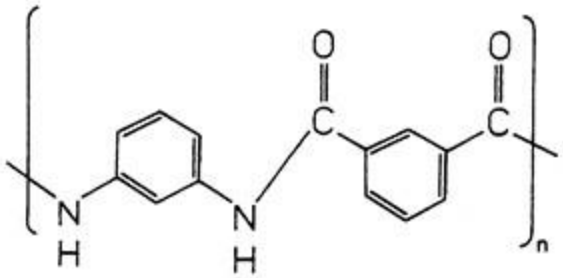
- **Nomex®** $(C_{14}H_{12}O_2N_2)_n$

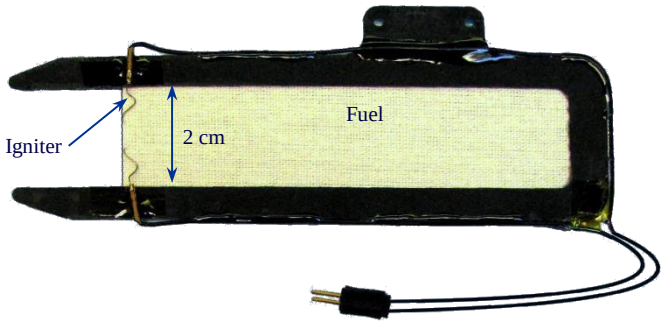
  HT90-40 fabric is a 12 mil thick fire retarded aromatic nylon ***fabric*** which does not melt or drip as it burns. When exposed to a heat source, the Nomex fibers swell and seal the spaces between the fibers, stopping air movement through the fabric and thus inhibiting heat transfer through the fabric.

  ULOI: 23.5% O_2

  Nomex-III

  ULOI: 22% O_2






Test D-08-12
September 2, 2009
Ultem-10mil
21.0%, 10.2 psia: Poor ignition
Test D-08-13
September 9, 2009
Mylar G-5.3mil + Kimwipe strip
15.6%, 10.2 psia: Burned
Test D-08-14
September 10, 2009
Nomex HT90-40
19.9%, 14.7 psia: EXTINCTION
Test D-08-15
September 15, 2009
Mylar G-5.3mil + Kimwipe strip
14.1%, 10.2 psia: EXTINCTION
Test D-08-16
September 17, 2009
Ultem-10mil
21%, 10.2 psia: Burned
Test D-08-17
September 22, 2009
Ultem-10mil
19.9%, 10.2 psia: EXTINCTION

Results:

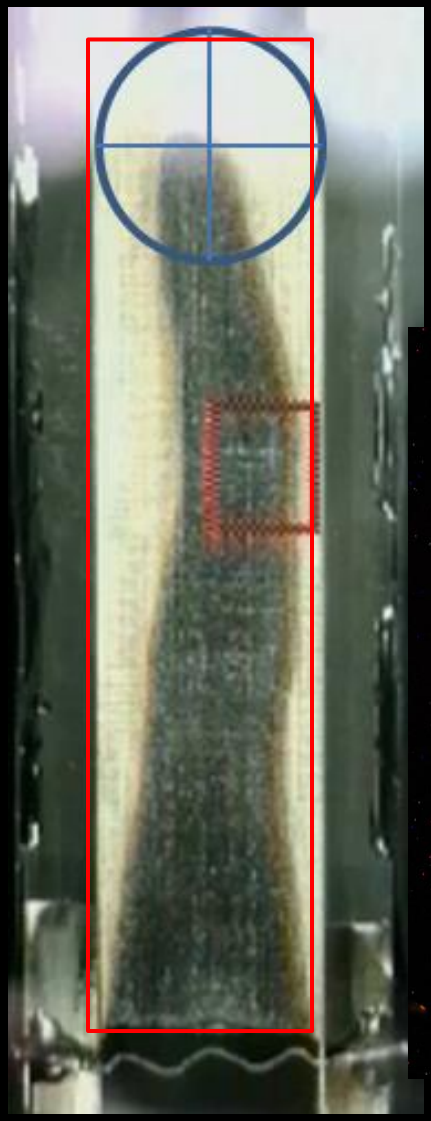
Ultem and Nomex HT90-40 samples did not burn on ISS atmosphere

- 21% to 22% O_2
- Flow speeds around 15 cm/s
- Similar to 1-g (Note: samples narrower on ISS)

Nomex-III sample did burn

NOMEX – III, 22% O_2, 1 atm 14 cm/s (0g)

2 cm wide sample



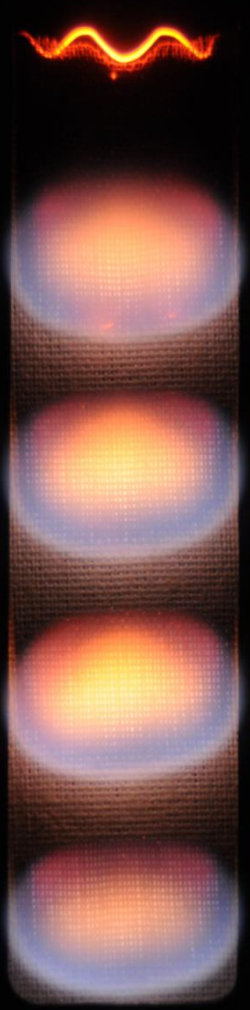


At 1g ULOI, but narrower sample

Nomex III sample residue after nearly complete concurrent propagation at 15 cm/s on ISS. Flame was shrinking in width until it extinguished within 1 cm of the end of the sample, as its width became smaller than ~ 1 cm, which is on the scale of a fingering flamelet.

Flow

Comparison of 0-g and 1-g Opposed Flow Flames

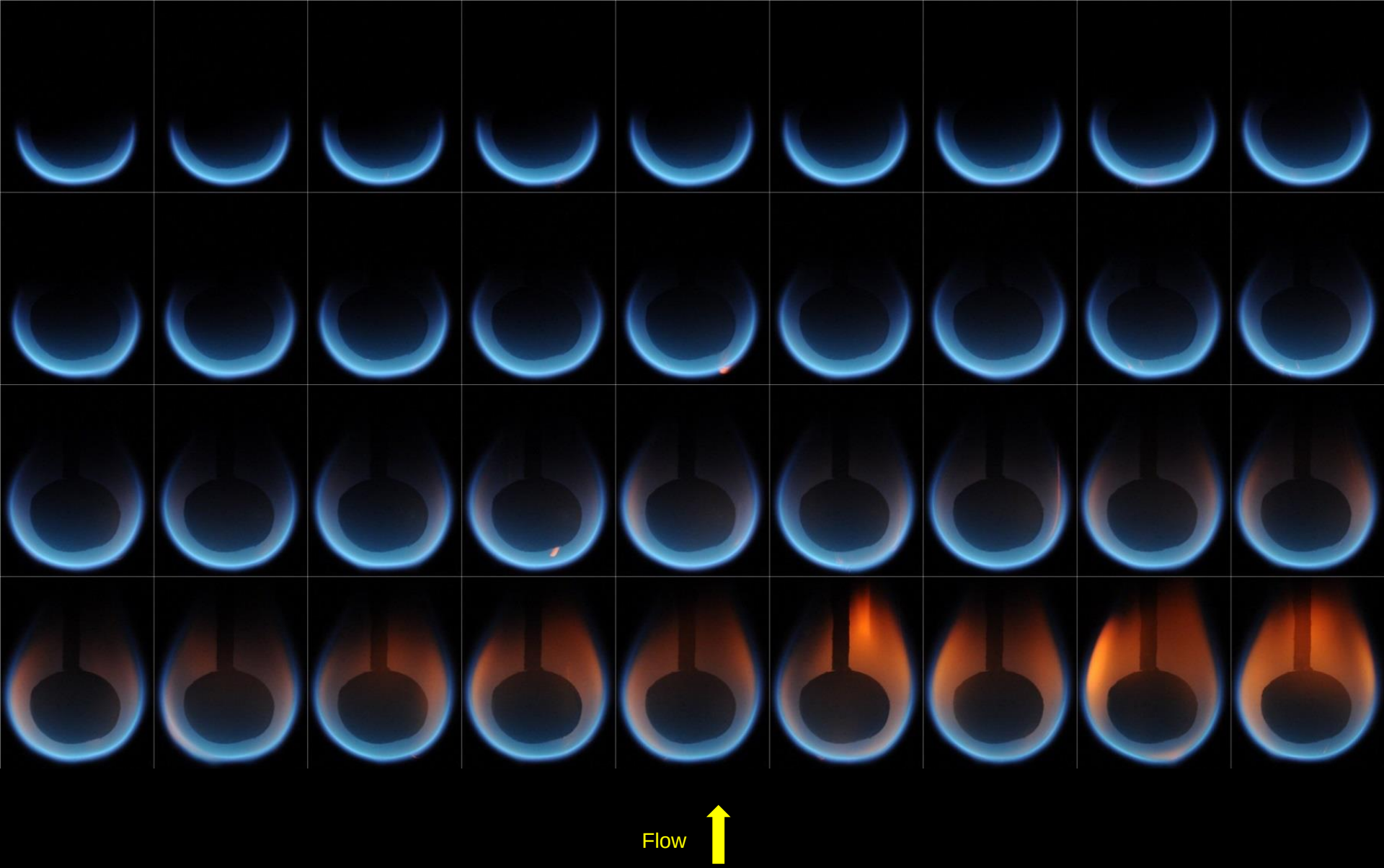




0-g with flow (10 cm/s)
Steady flame size and spread
Convex base
Extended blue zone

1-g
Flame does not propagate downward

3. Spherical PMMA samples



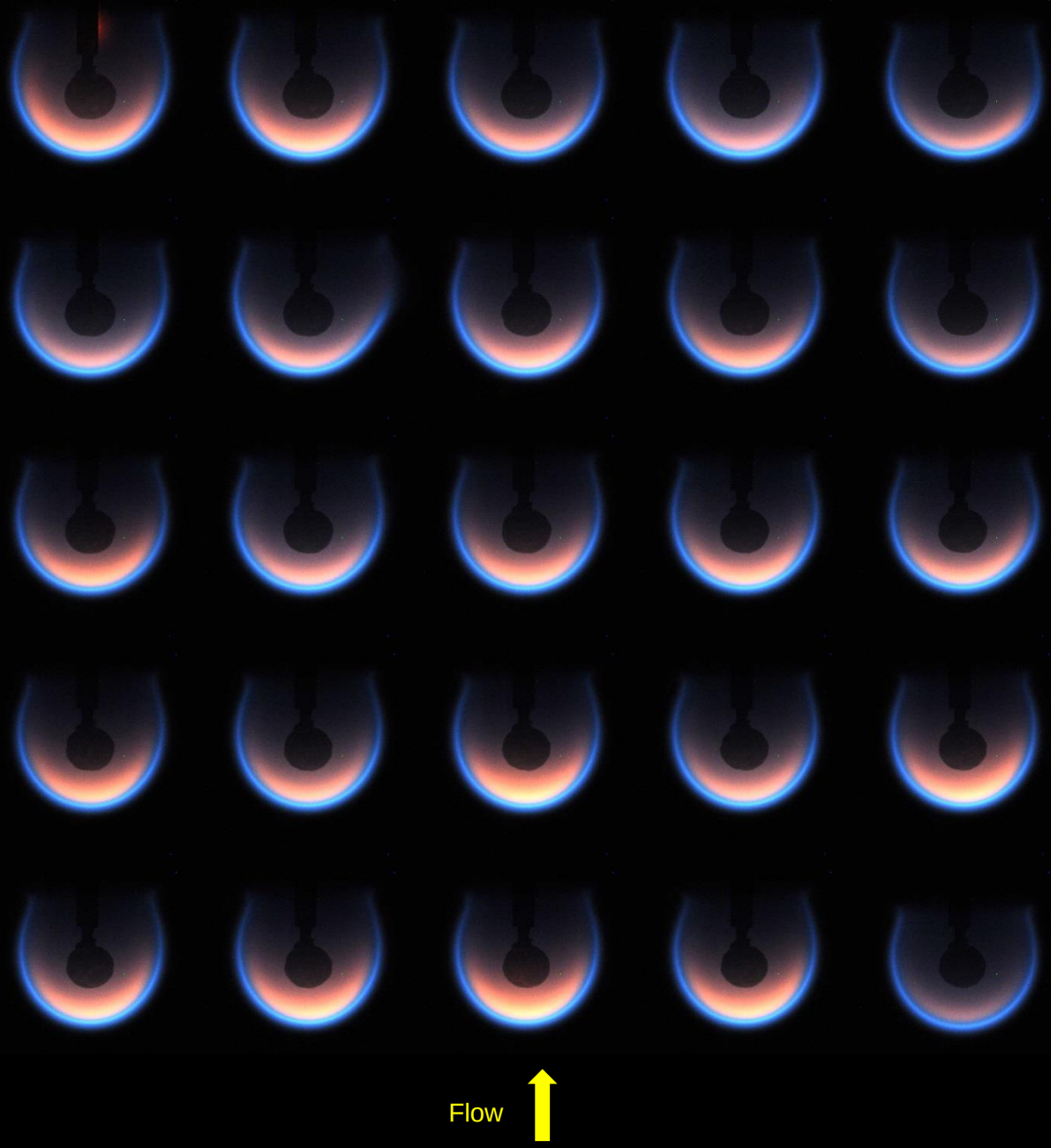
Flame sequence of a burning 2-cm diameter PMMA sphere at 17% oxygen and 12 cm/s flow. The images are about 1.5 sec. apart. The high resolution images allow model comparison of flame growth rate, flame-to-fuel distance, and the solid regression rate.



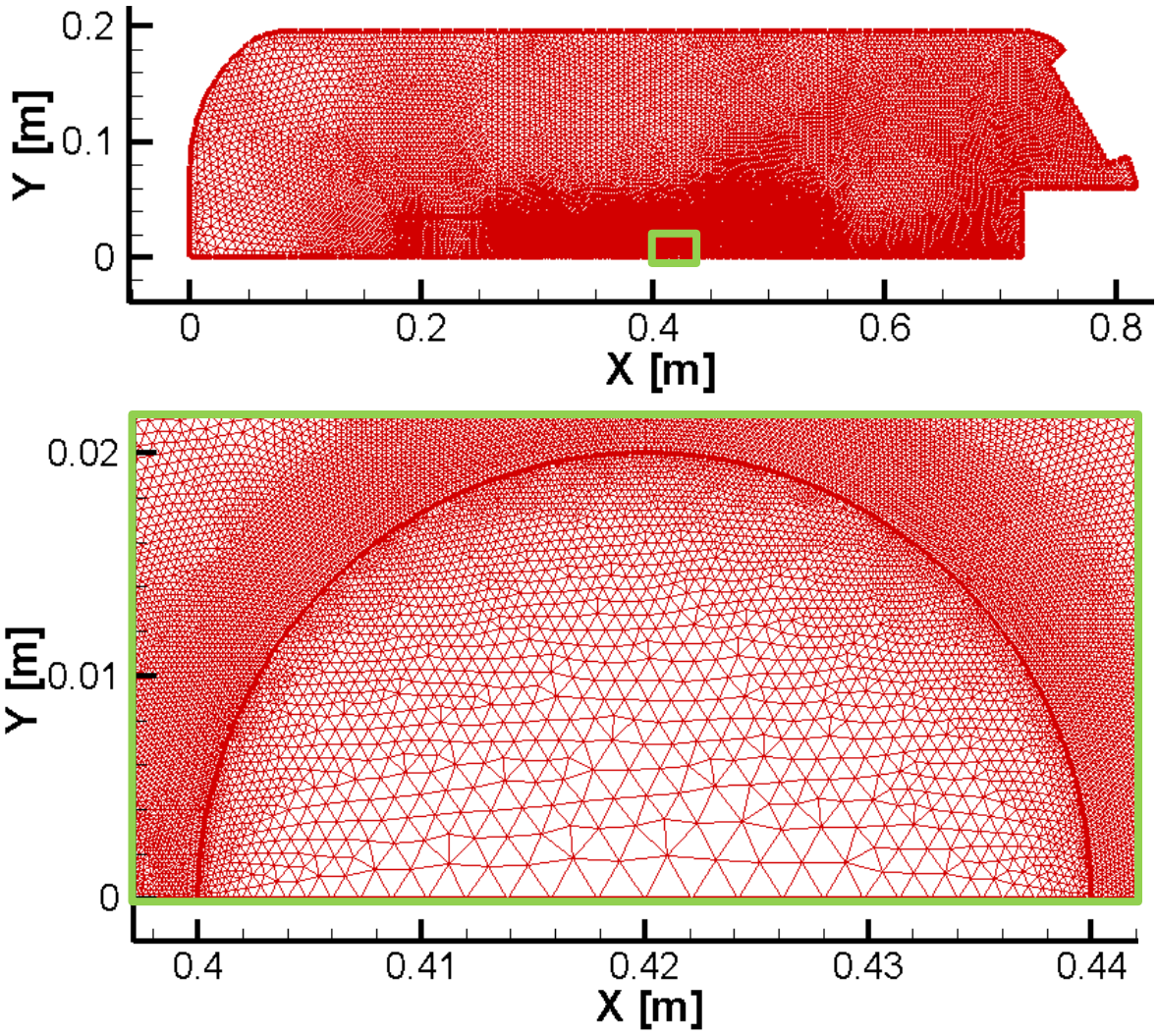
Flame sequence of a burning PMMA sphere (1-cm diameter) at 17% oxygen and less than 1 cm/s.

(Images are 1.3 sec. apart)

The fuel itself is clearly visible in this contrast-enhanced montage. This enables us to get an accurate measure of the fuel burning rate which is an important parameter to characterize the system for comparison to the model.



Mesh

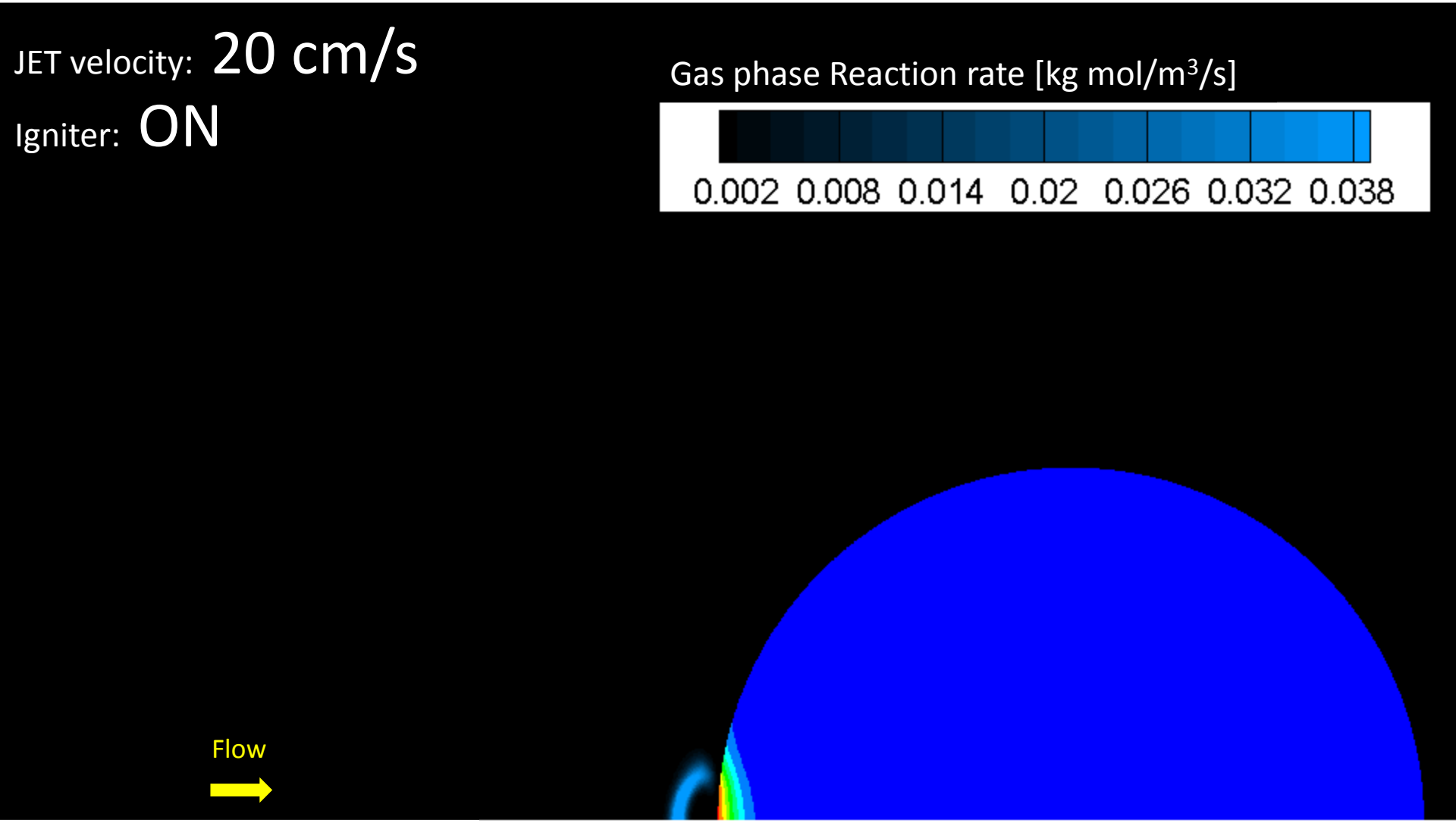


Having 33930 points total, Using quad-core CPU (4 processors) with 16GB memory, it takes 2.7 hours to compute one second in the computation (with capability of shape change due to solid surface regression).

Flame spread (10sec.)

JET velocity: 20 cm/s

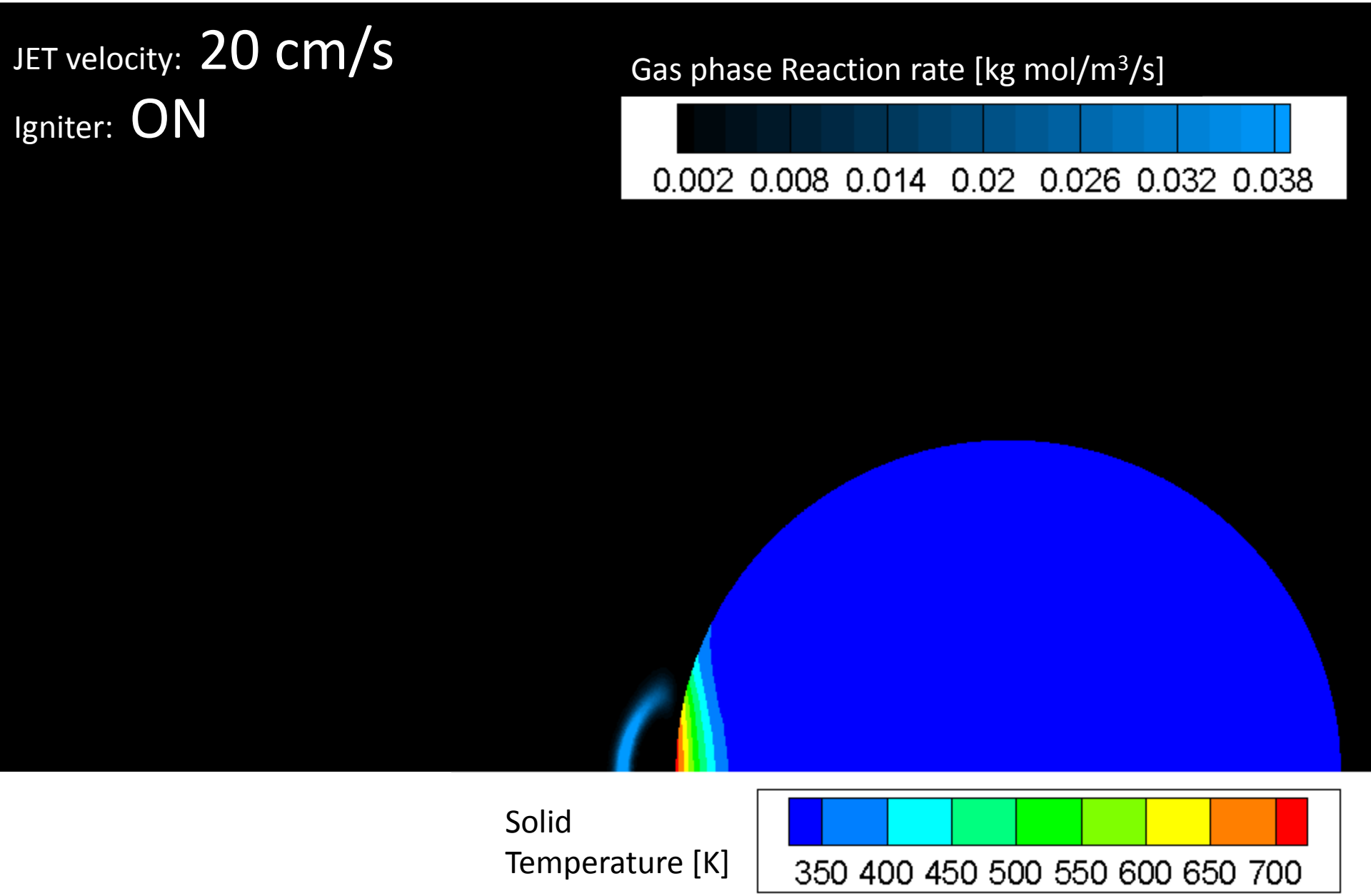
Igniter: ON



Flame spread (20sec.)

JET velocity: 20 cm/s

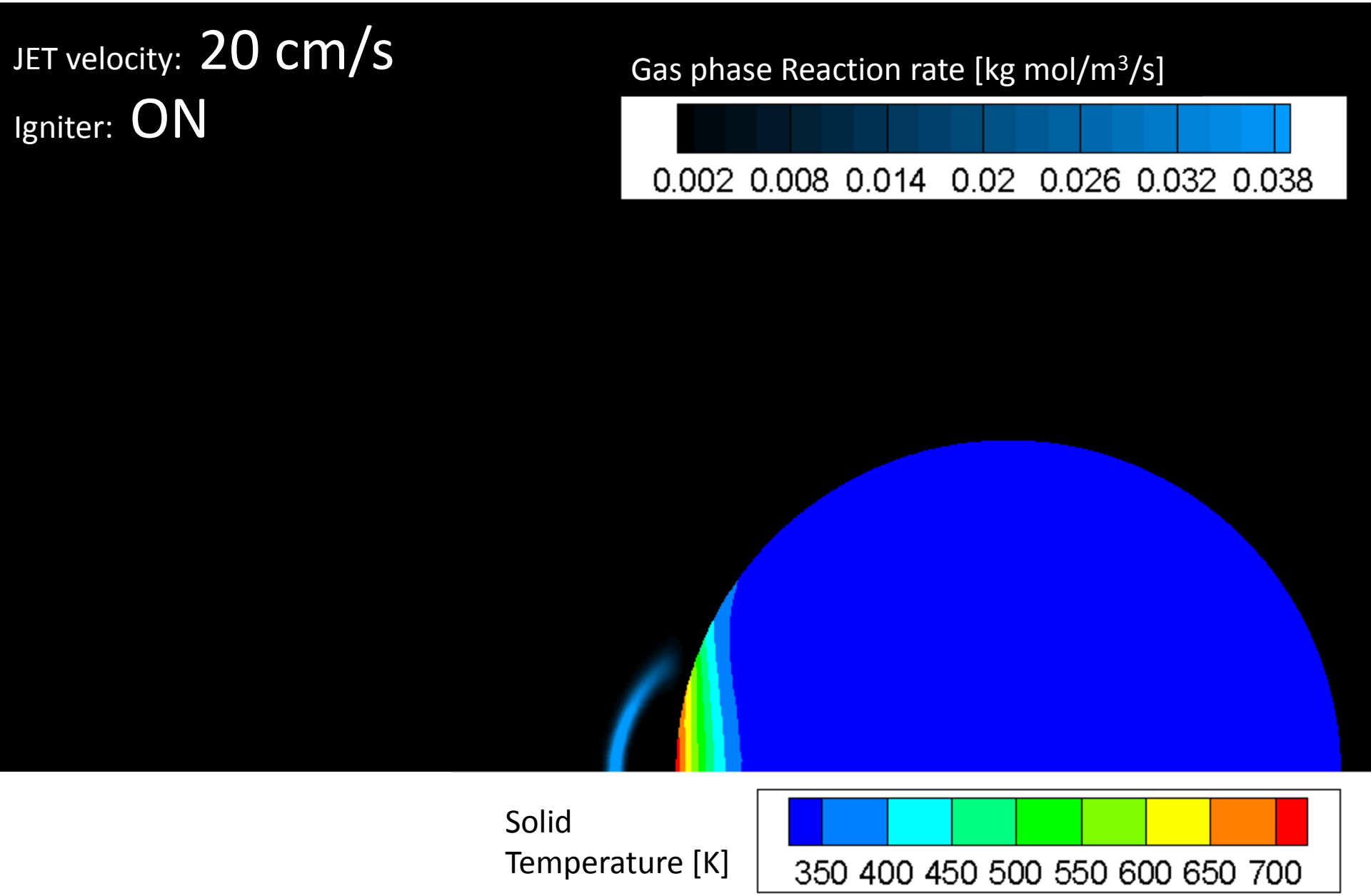
Igniter: ON



Flame spread (30sec.)

JET velocity: 20 cm/s

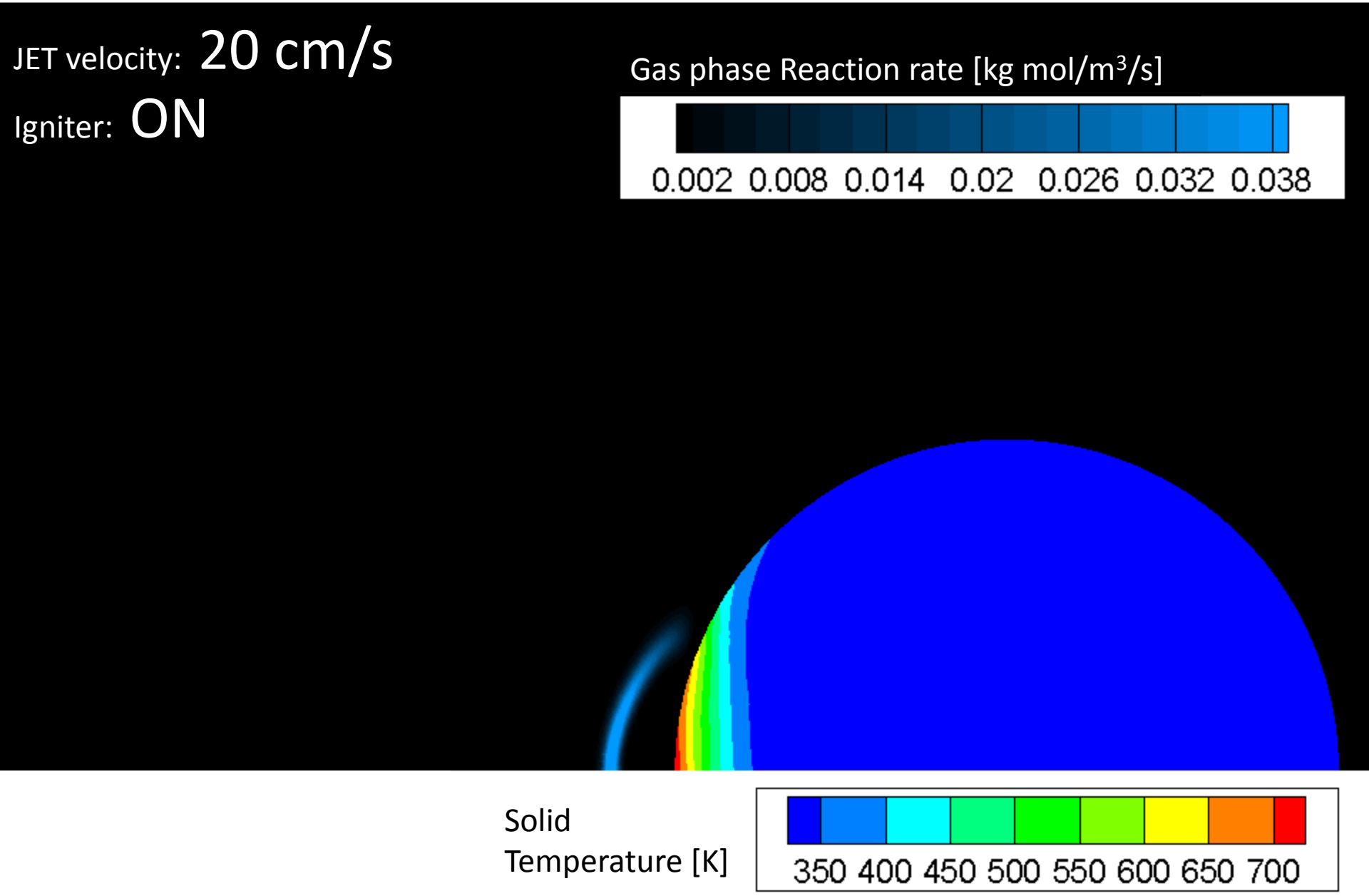
Igniter: ON



Flame spread (40sec.)

JET velocity: 20 cm/s

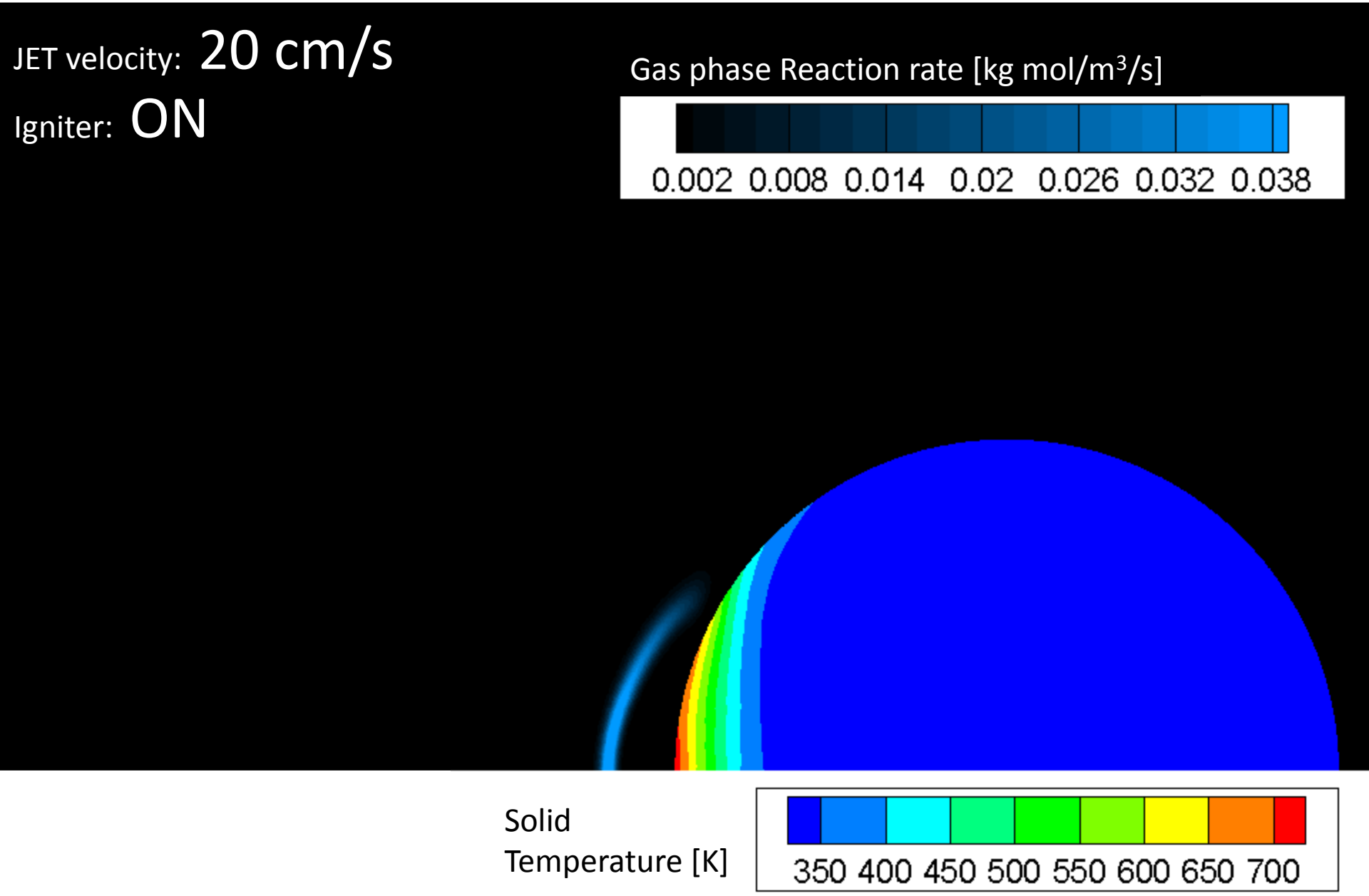
Igniter: ON



Flame spread (50sec.)

JET velocity: 20 cm/s

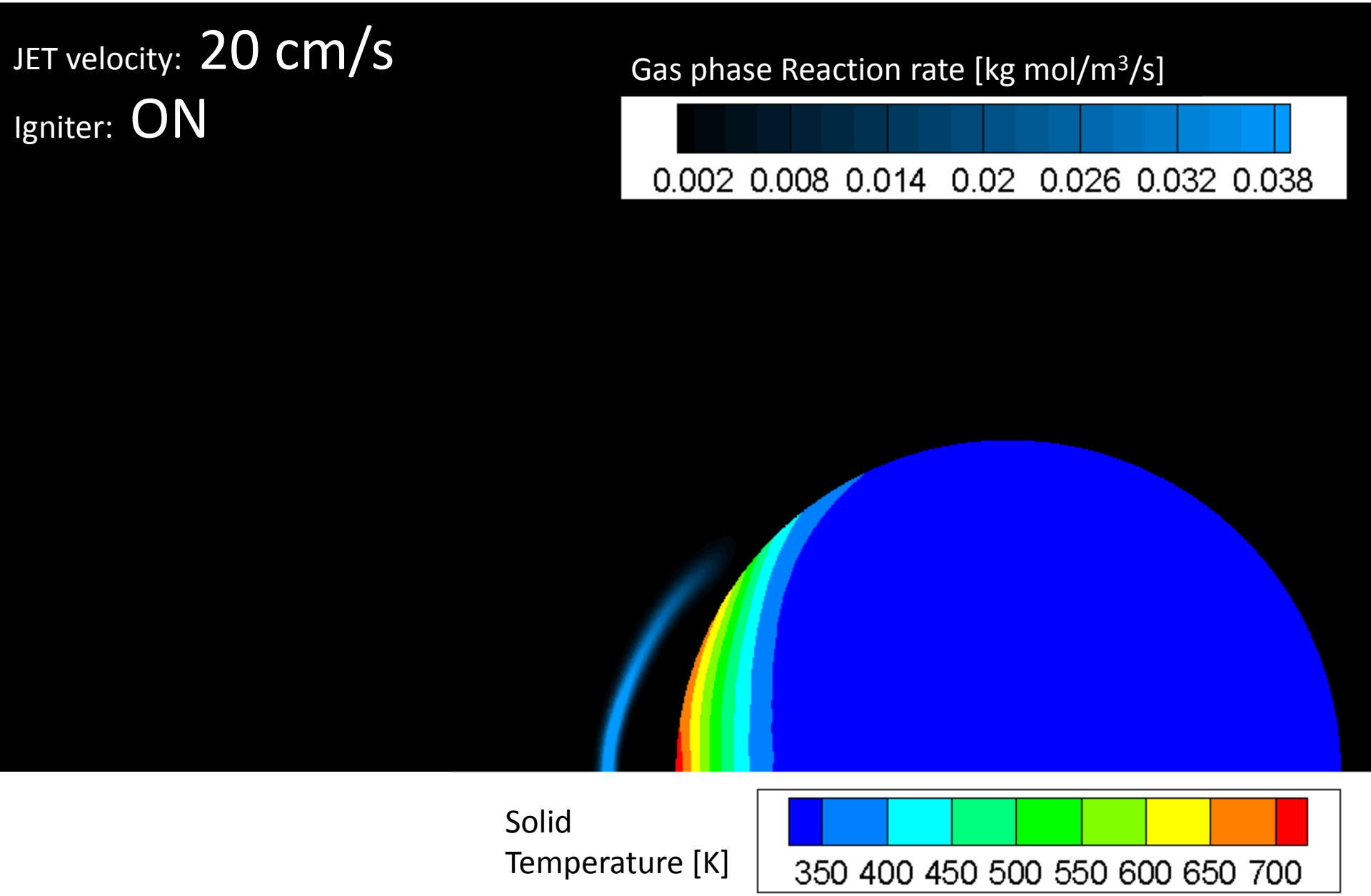
Igniter: ON



Flame spread (60sec.)

JET velocity: 20 cm/s

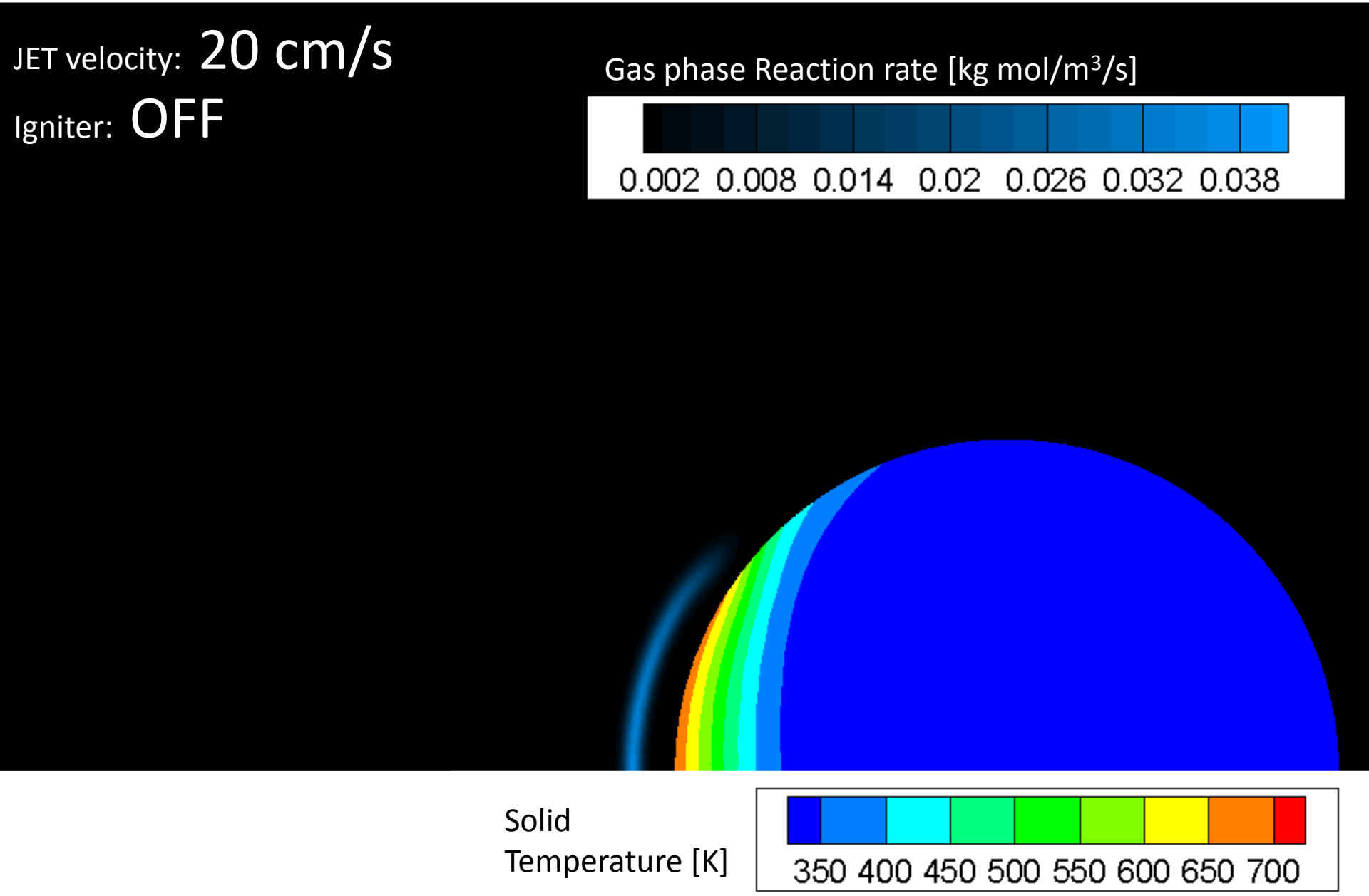
Igniter: ON



Flame spread (70sec.)

JET velocity: 20 cm/s

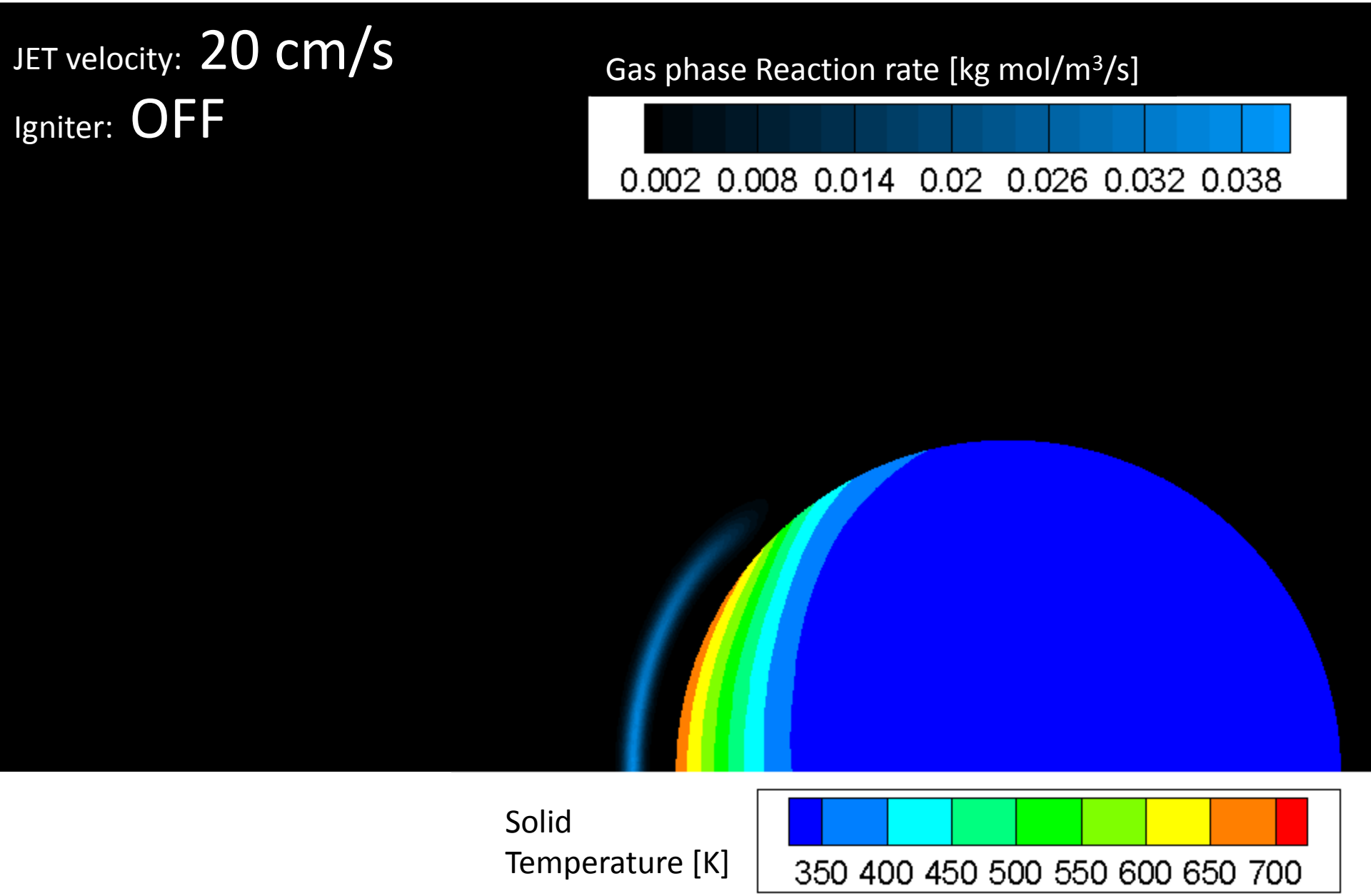
Igniter: OFF



Flame spread (80sec.)

JET velocity: 20 cm/s

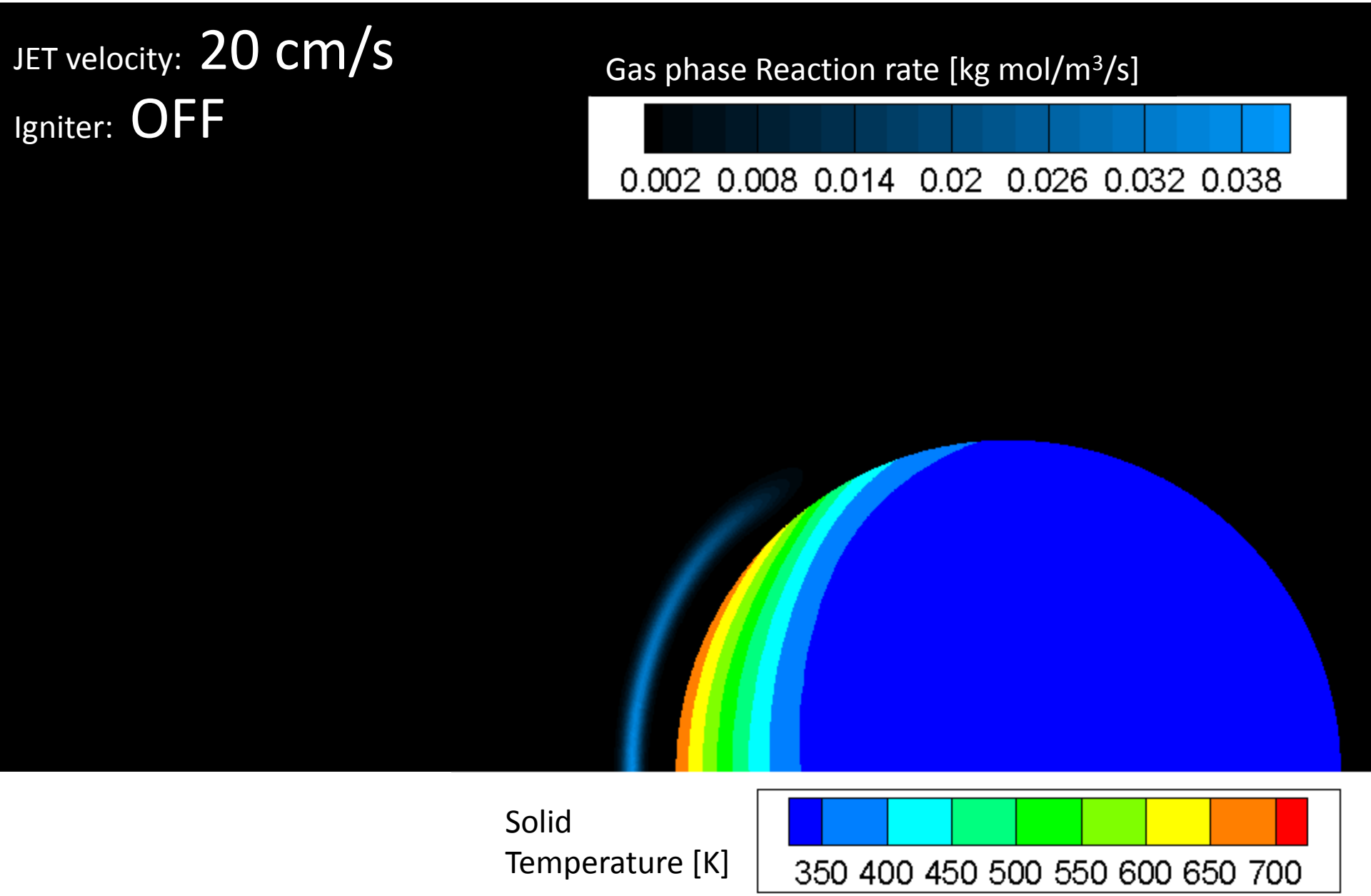
Igniter: OFF



Flame spread (90sec.)

JET velocity: 20 cm/s

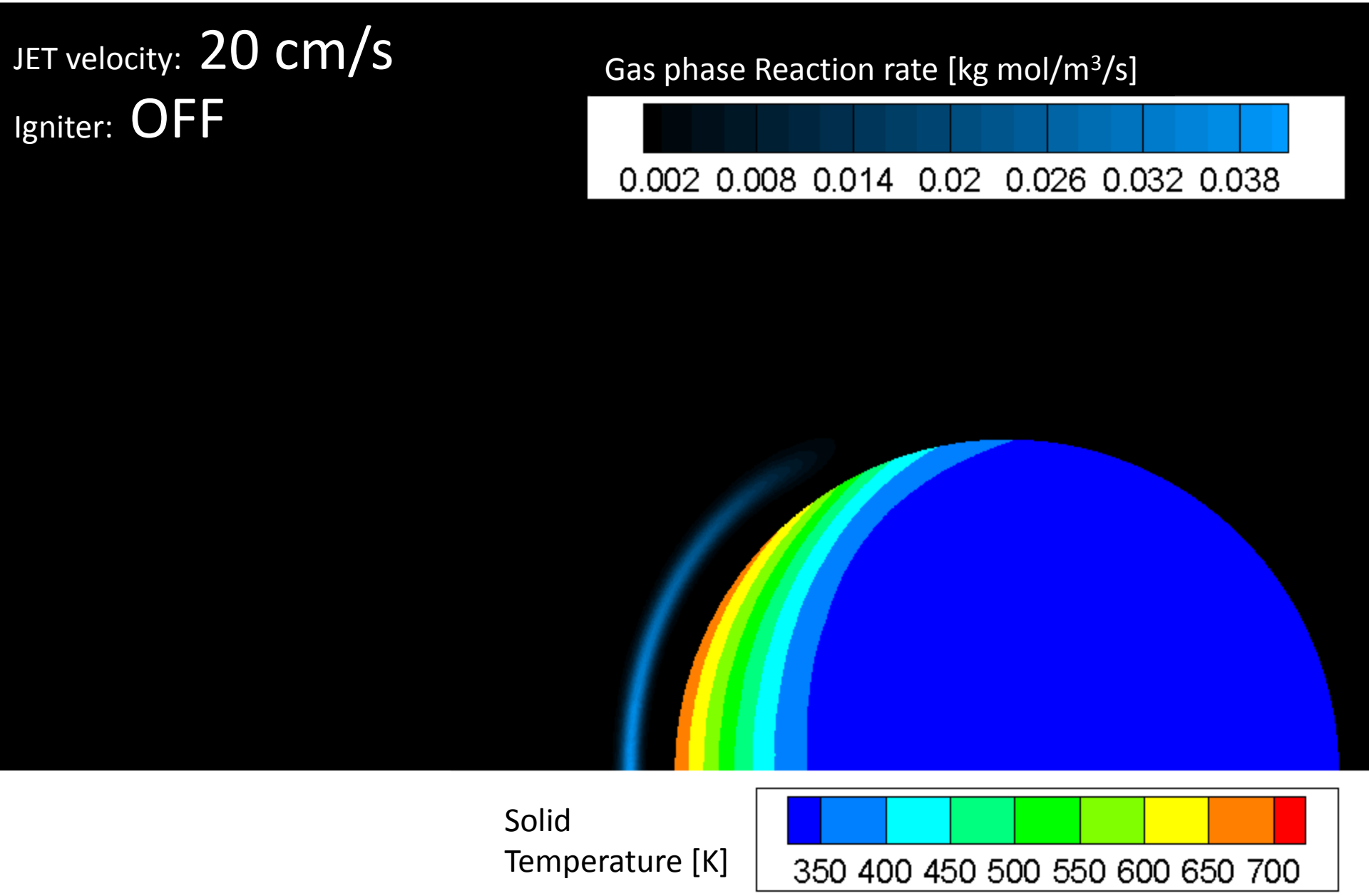
Igniter: OFF



Flame spread (100sec.)

JET velocity: 20 cm/s

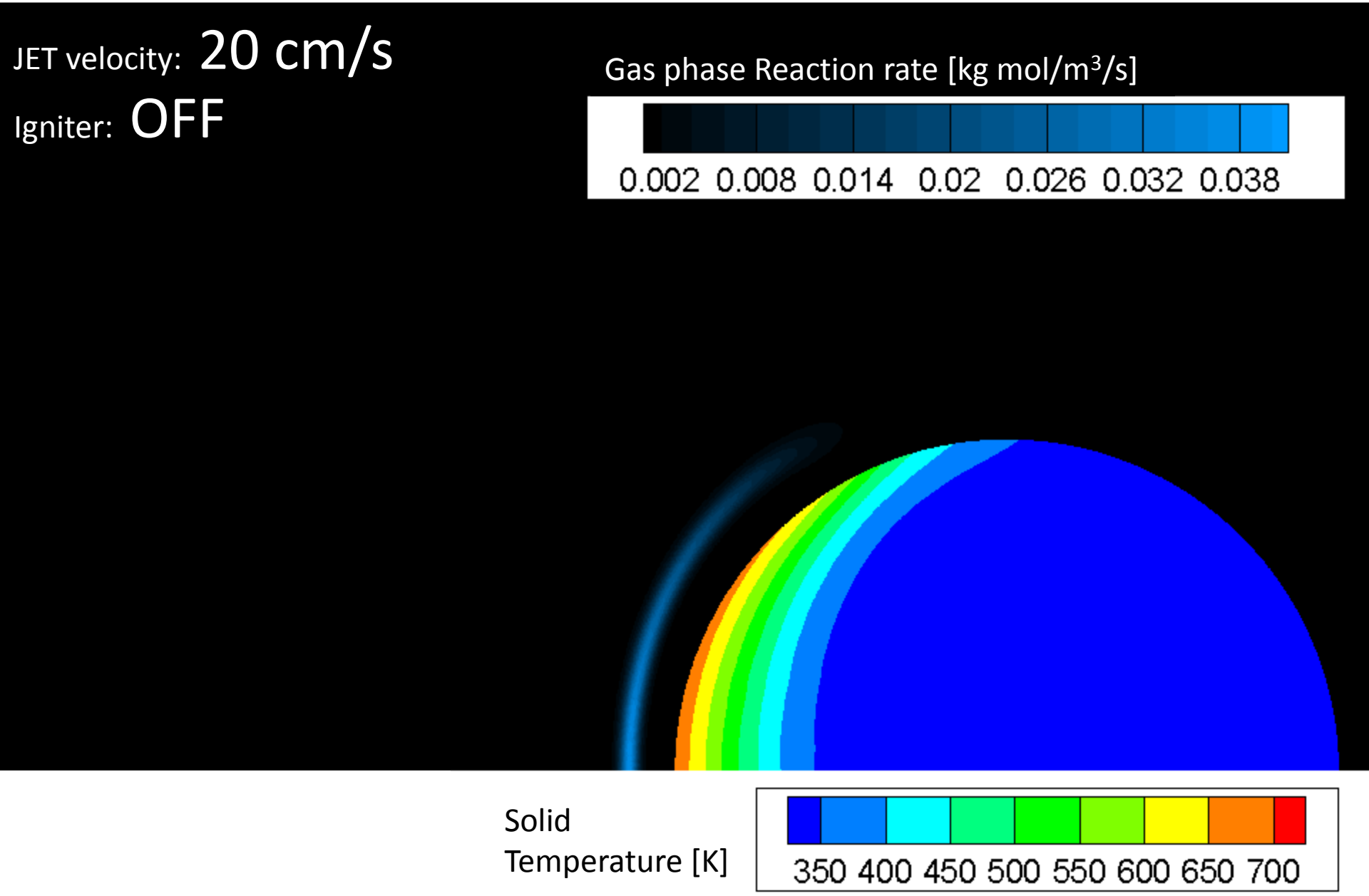
Igniter: OFF



Flame spread (110sec.)

JET velocity: 20 cm/s

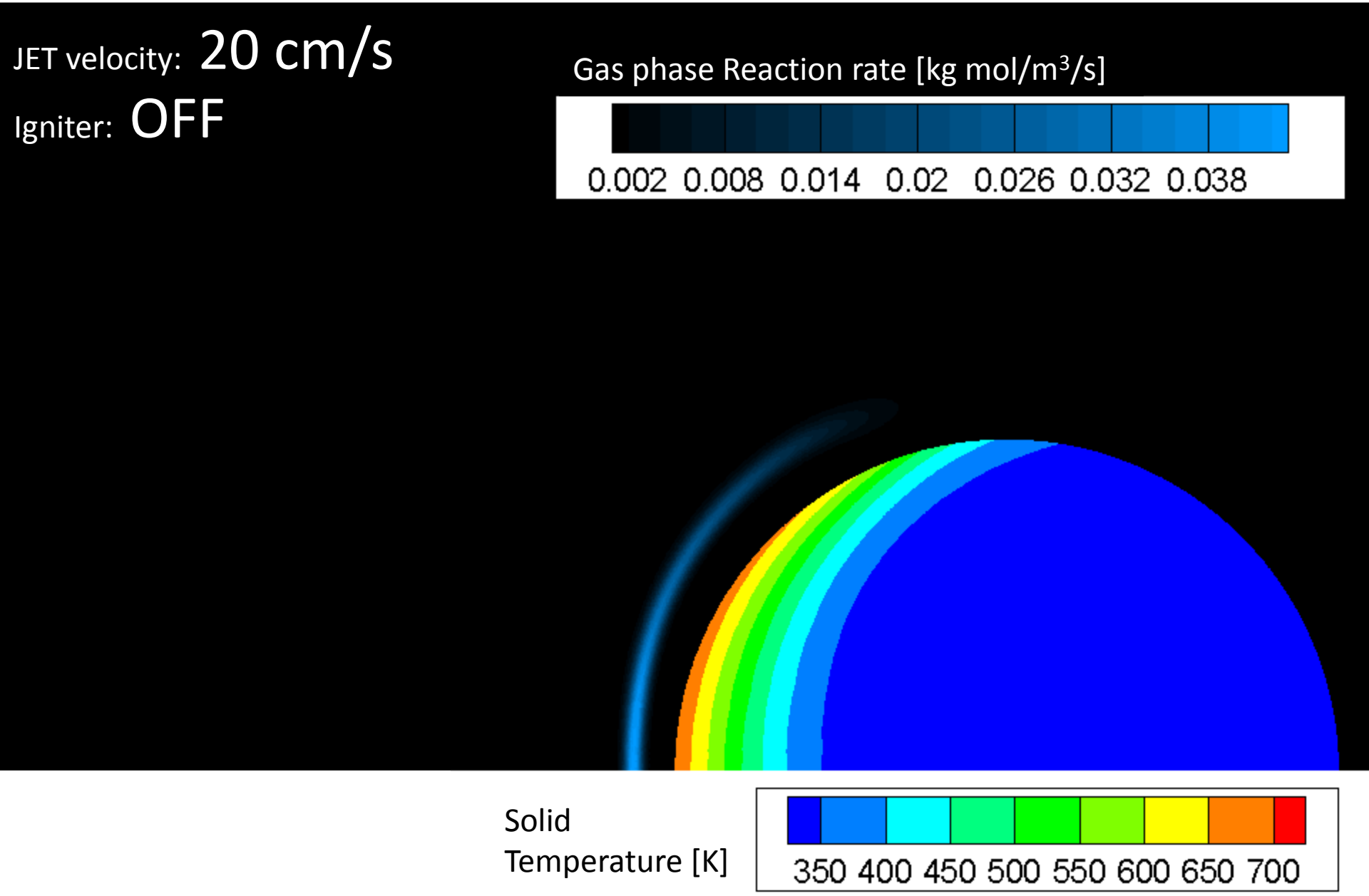
Igniter: OFF



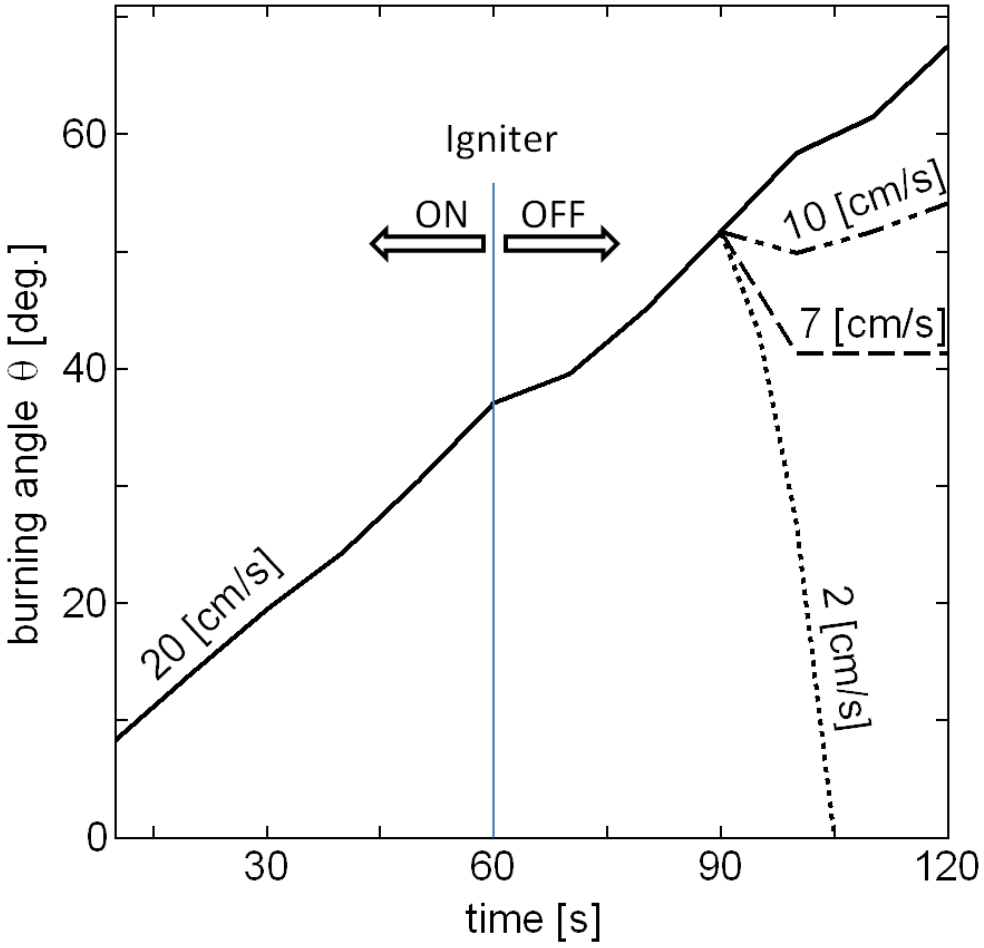
Flame spread (120sec.)

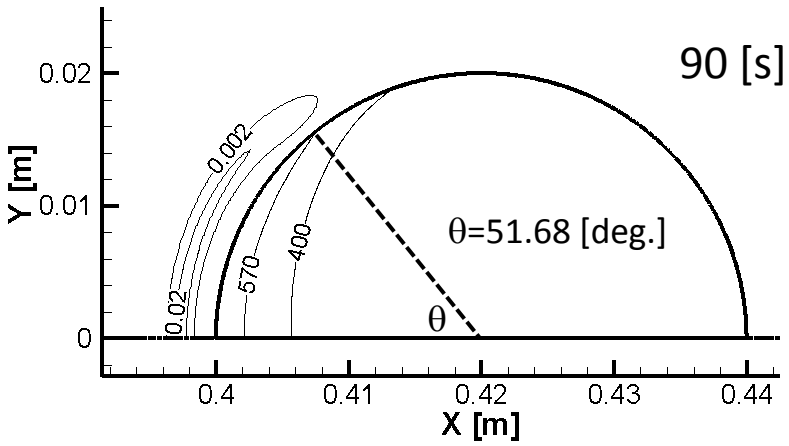
JET velocity: 20 cm/s

Igniter: OFF

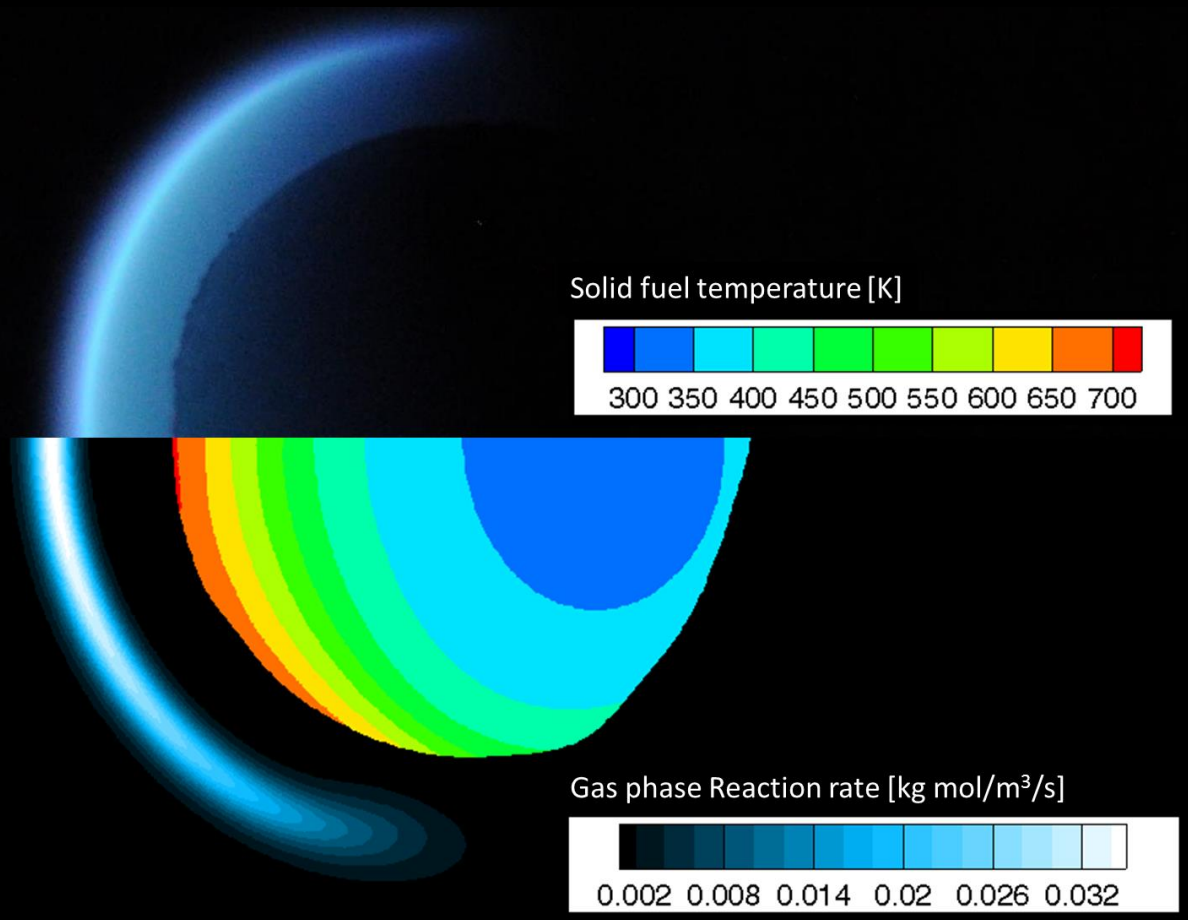


Flame spread graph





Burning angle θ is defined by the solid surface temperature (570K).



Flame burning an acrylic sphere (top) from is compared to model computation (bottom). The comprehensive model even includes solid phase shape change effects. Adjusted solid phase chemical kinetic parameters will improve the prediction of the flame standoff distance.

Fuel: Acrylic sphere; Atmosphere: 17% O_2/N_2; 1 atm; Flow speed: 12 cm/s (left to right)

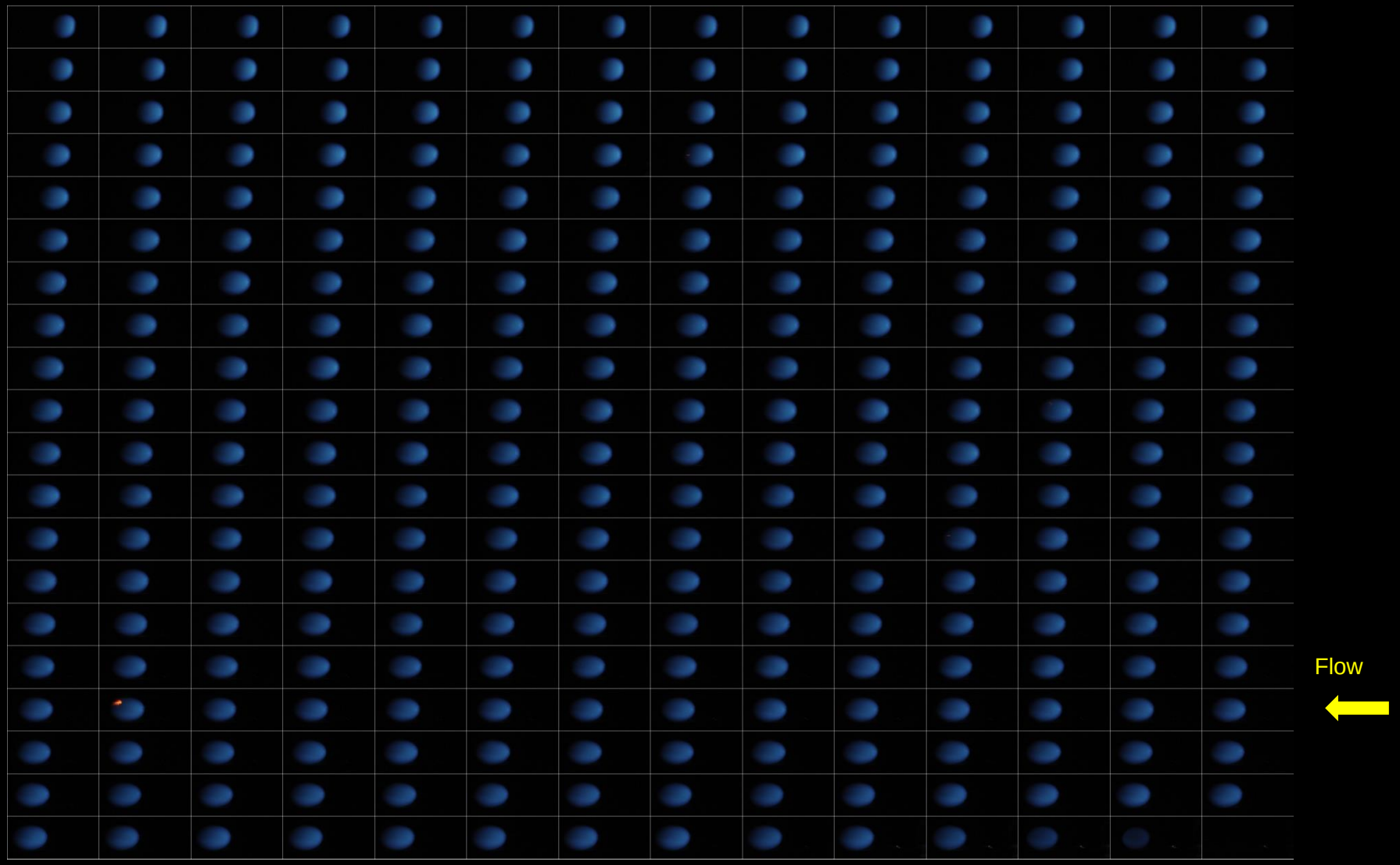
Dynamic flow change effects



Flow

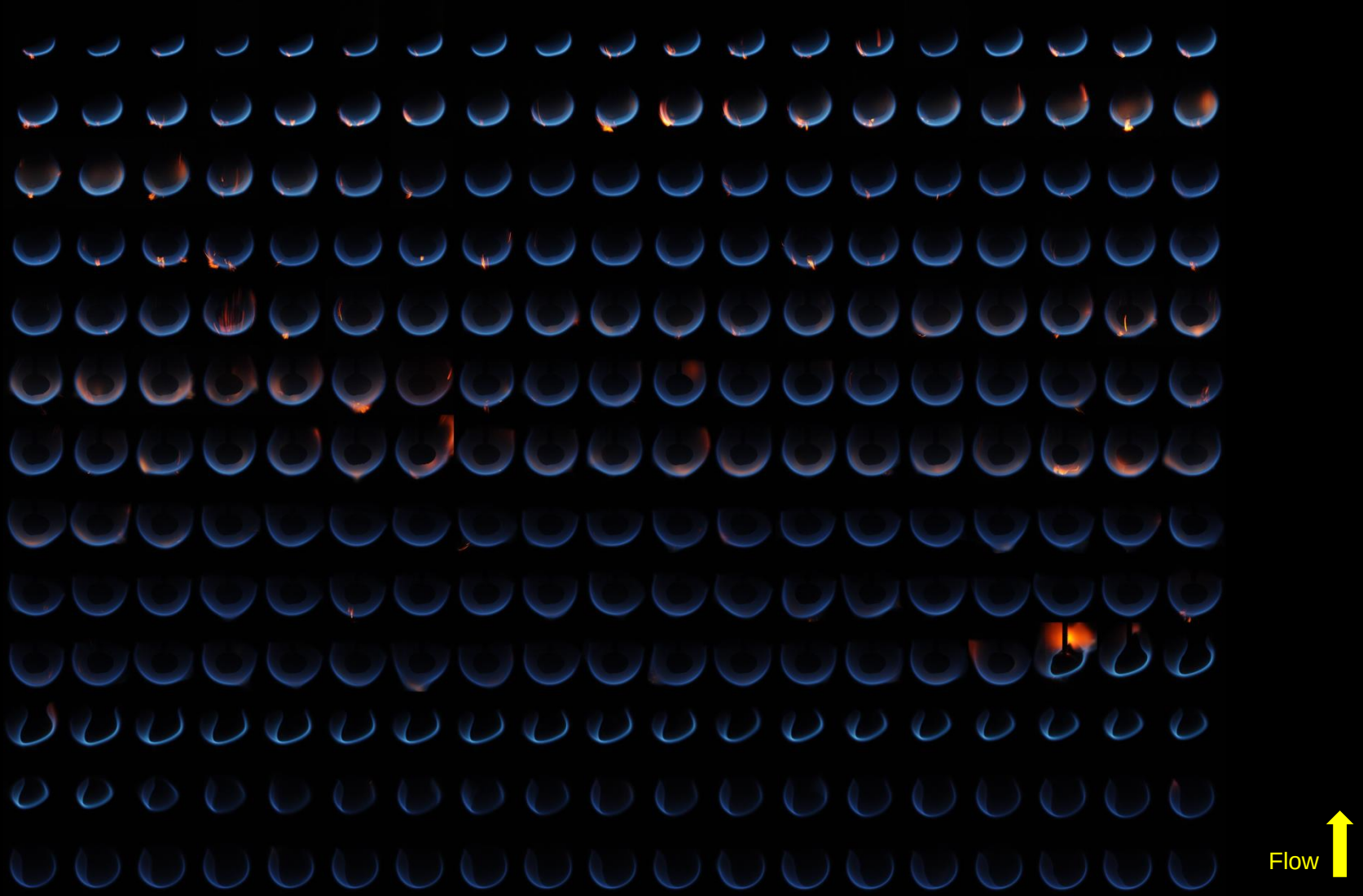
4. Flat PMMA Samples





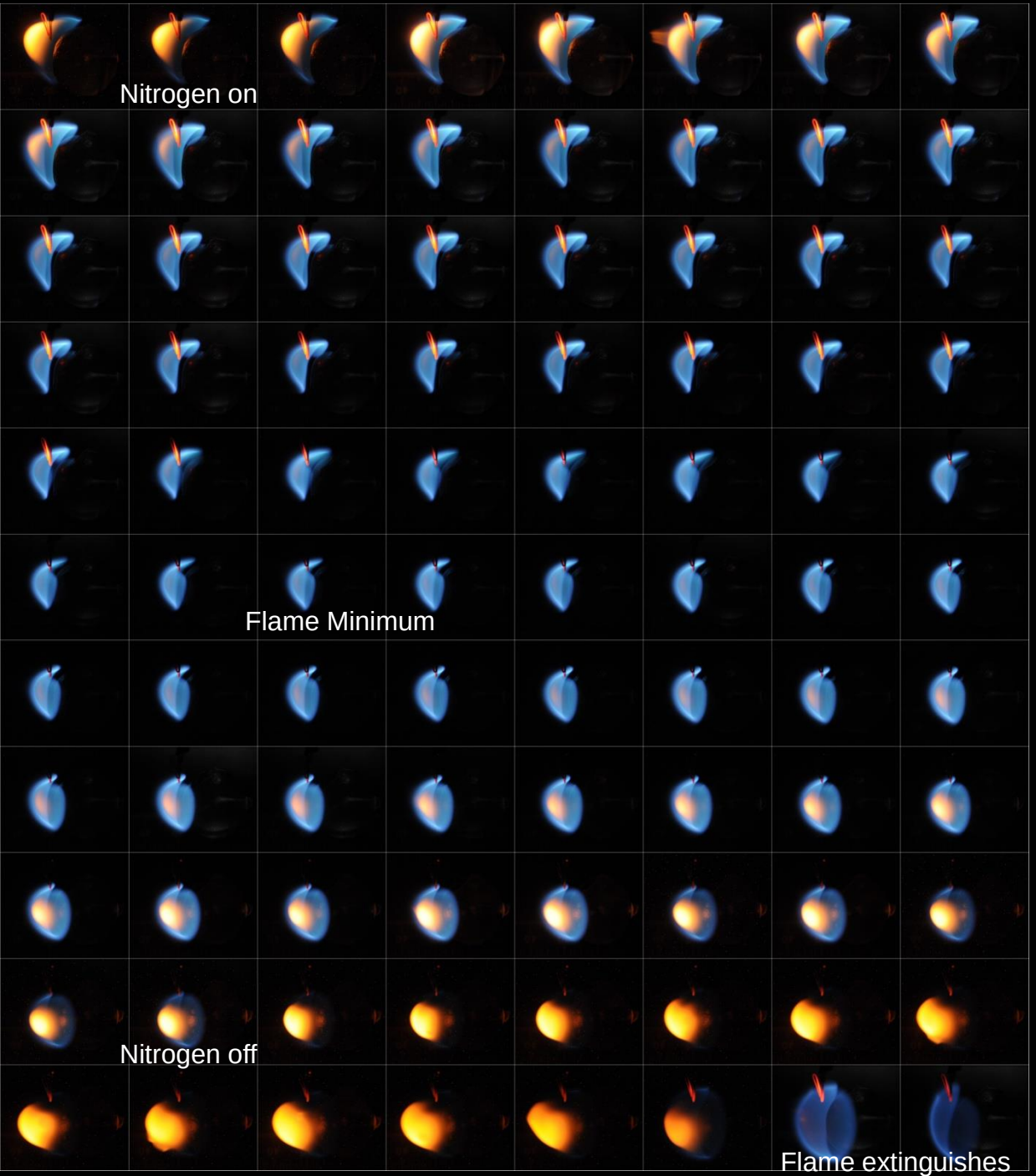
Top view image montage for a 1-cm wide flat acrylic sample burning in air at low flow speed (around 9 cm/s). Images progress from left to right then top to bottom (1.3 seconds between images). The air flow direction is from right to left. The flame is about 1 cm in size and can persist for a very long time at this low air flow speed. The flame goes out only in the last two frames when the air flow is completely shut off.

5. Nitrogen Suppression and Wake Flames



Flow of less than 1 cm/s. Near the end of the burn, a jet of nitrogen is imposed but it fails to extinguish the flame. Finally after a series of flow reductions, the flame goes out when the flow speed is zero.

Fuel: PMMA; 15-mm initial diameter; 17% O_2/N_2; 1 atm; Time between images: 1.3 sec



Flow

PMMA sphere burning in wake configuration in air.

Flow is from right to left

Nitrogen flow of 500 cc/min weakens but does not extinguish the flame

6. Candles



Image sequence of a burning paraffin wax candle within a ceramic tube at 17% oxygen and less than 1 cm/s flow. Images are about 9 sec. apart and the flow direction is up. The test is concluded when the flow is turned off and the flame rapidly goes out.

Flow

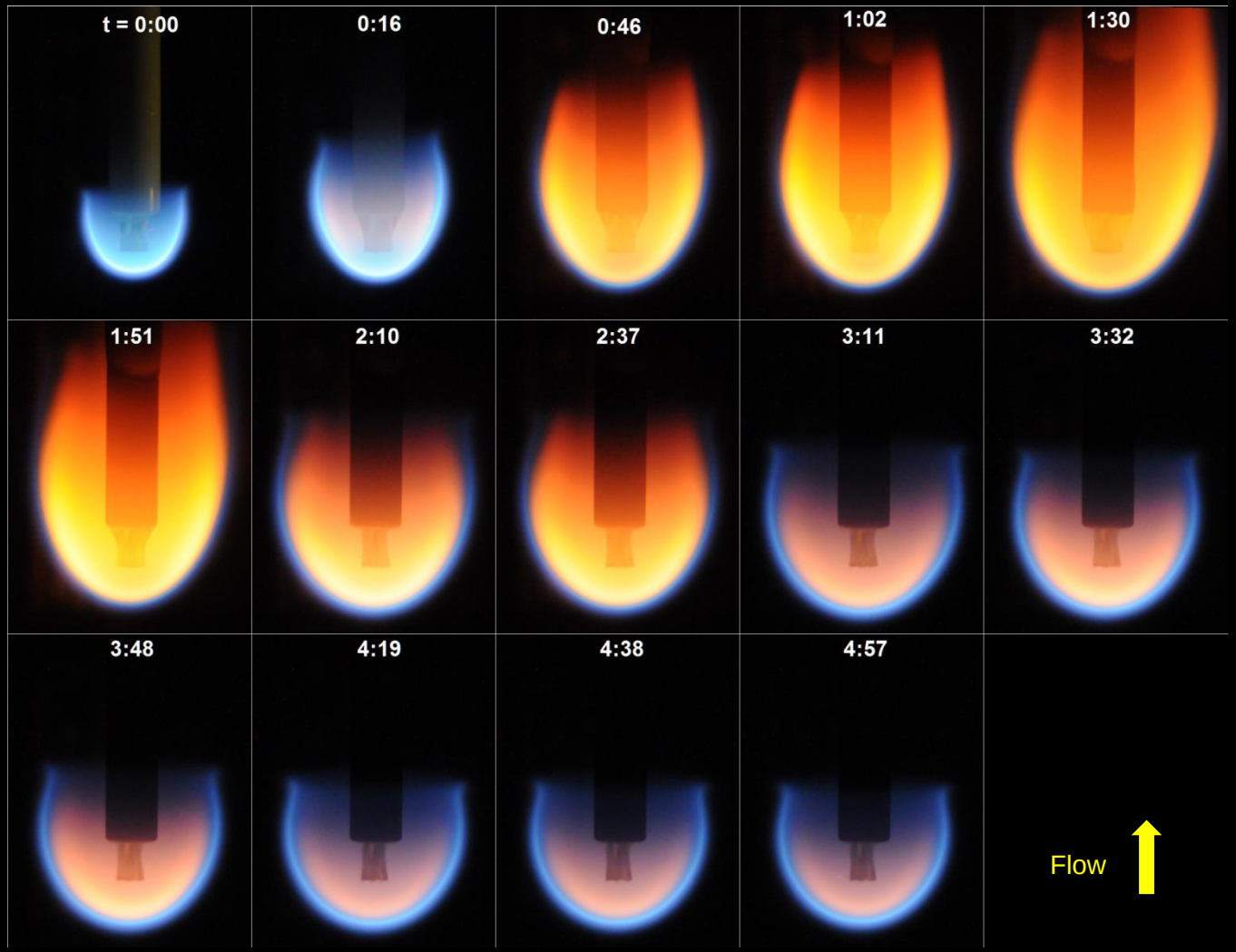
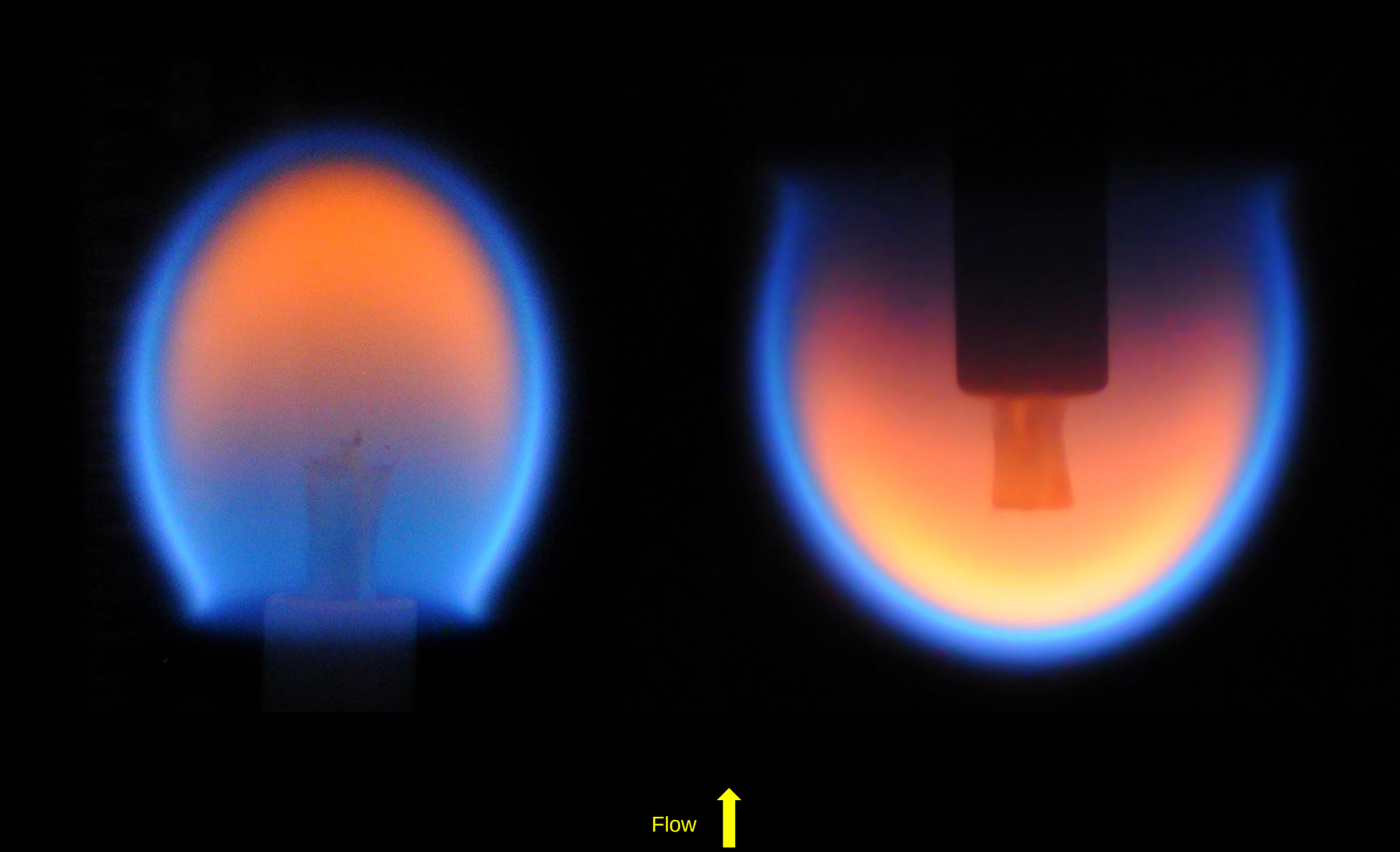


Image sequence of a burning candle in microgravity. Flow is from bottom to top.

Atmosphere: 17% O_2/N_2; 1 atm; Flow speed: Varied --> 5, 4, 3, 2, 1, <1 cm/s;
Relative time (mm:ss) is indicated on the images



Flow direction comparison on candle flame burning in 17% O_2 flowing at around 1 cm/s.

Flow is from bottom to top.



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Carol Reynolds, NASA Marshall Space Flight Center

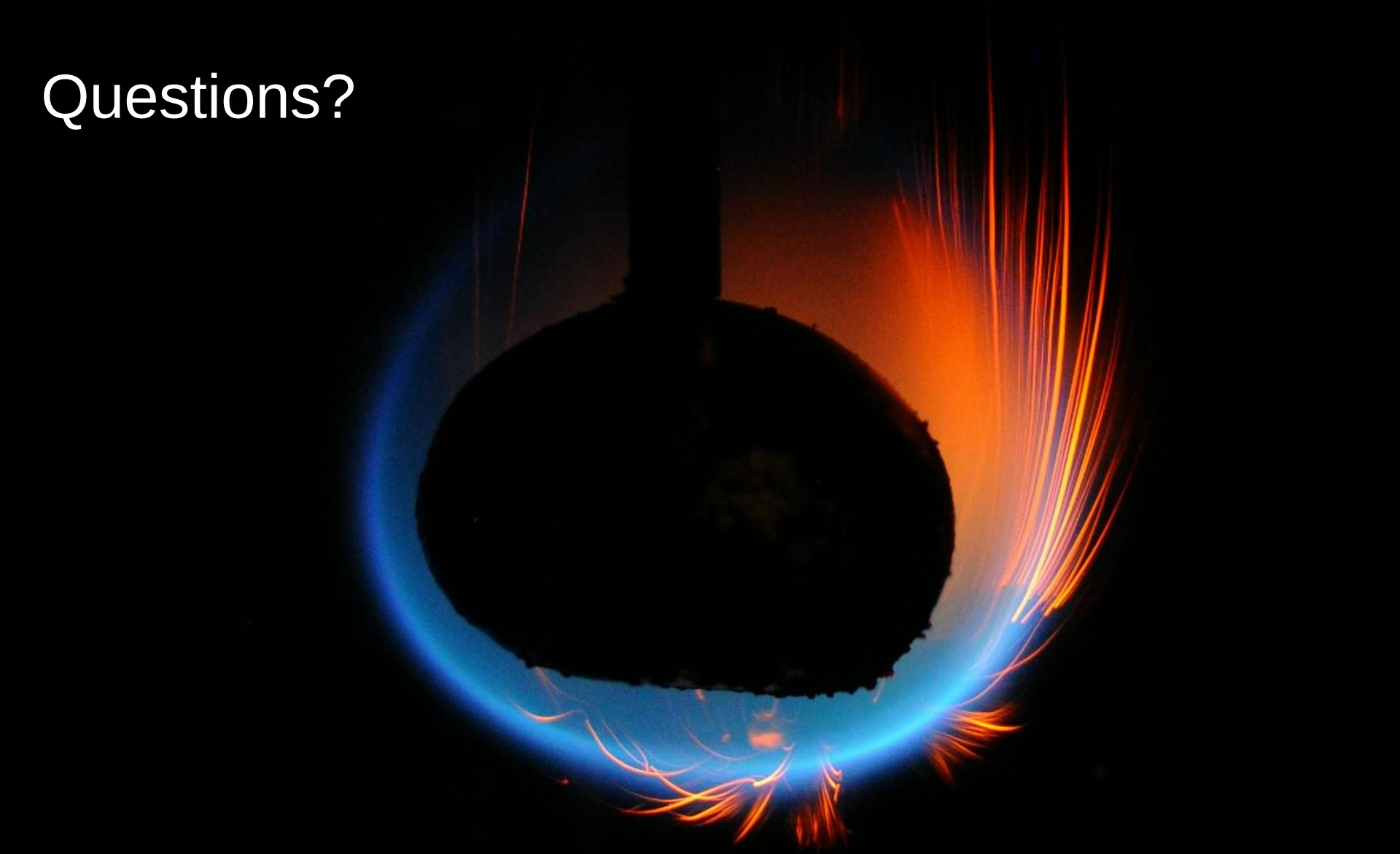
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Joe Acaba, NASA Johnson Space Center

Suni Williams,NASA Johnson Space Center

Chris Cassidy, NASA Johnson Space Center

Questions?



PMMA sphere burning in 17% O_2 with occasional vapor jetting